6DATA007W – Final Project Report
"SECTORAL IMPACT ANALYSIS OF THE COVID-19 PANDEMIC IN THE UK."

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Declaration

This report has been prepared based on my own work. Where other published and unpublished source materials have been used, these have been acknowledged in references.

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Use of Generative AI

In this assessment, I took help of **Gemini AI** in **Google Colab** and also used **OpenAI's ChatGPT** to assist in the adjustment of the written components of this final report, based on my own analysis, code, data visualisations, and notebooks. It was used to formulate the sentences. The tool supported structuring academic language and summarising statistical findings. All original code, data handling, and analytical insights were completed independently.

Abstract

The COVID-19 pandemic emerged as one of the most significant global disruptions in recent history, affecting economies, supply chains, and labour markets on an unprecedented scale. In the United Kingdom, businesses were forced to adapt to quickly changing conditions, ranging from lockdown-induced closures to labour shortages and international trade disruptions. While some sectors managed to remain resilient, others, particularly those reliant on in person interactions, had significant operational and financial setbacks. This research looks on how the pandemic has affected five important industrial sectors in the UK: **Accommodation and Food Services (Hospitality), Wholesale and Retail Trade (Retail), Manufacturing, Transportation and Storage, and Information and Communication**.

This project's main goal was to investigate how unexpected events like a pandemic, impact the industrial environment in several important areas. Analysing the changes in each sector's financial performance, workforce dynamics, trade activities, and strategic adaptations during the peak of the COVID-19 pandemic and in the months that followed was part of this. Depending on the dataset category, the study focused on the period from **July 2020 to late 2021** and used data that was only obtained from the **UK Office for National Statistics (ONS).**

A mixed-methods strategy was used, to find sector-specific trends, the investigation started with exploratory data analysis (EDA) using Python (Pandas, Seaborn, Matplotlib). Statistical hypothesis testing was then conducted to compare the early and late pandemic phases using independent t-tests. To facilitate cross-sector comparison, data obtained from 25 sector-specific files was combined into five merged datasets (Financial, Workforce, Import, Export, and Strategic). In order to enable users to examine visualisations, interact with the datasets, and download data and graphs for additional study, a Streamlit dashboard was created.

The results showed that, especially during the second lockdown, the hospitality and retail industries saw the biggest drops in employee turnover and the greatest furlough rates. Manufacturing and Information and Communication, on the other hand, demonstrated more financial performance resilience, although still having to deal with major workforce and supply chain issues. In terms of shortages in workers and trade operations, transportation and storage were particularly impacted. Significant disparities between early and late periods were found in the majority of sectors, according to statistical studies, particularly with relation to labour dynamics and financial performance. Retail and hospitality showed indications of a comeback by mid-2021, while industry

dropped behind, according to the visualisations, which also highlighted sector-specific recovery trajectories.

The study's conclusions indicate that resilience was significantly influenced by sector-specific factors, including business style, digital capacity, and rely on international trade. In addition, the data highlights the significance of effective government support and flexible business models in times of crisis. Policymakers, researchers, and business stakeholders can use the dynamic tool provided by the interactive dashboard developed as part of this project to examine the pandemic's long-term effects and get ready for any future unexpected disruptions.

Overall, the project demonstrates the value of data-driven analysis in understanding economic shocks and offers insights into how different sectors can build resilience against future global crises.

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1. Introduction

The COVID-19 pandemic has thrown a new threat to the global industry landscape in recent years, causing massive disruptions to trade operations, workforce dynamics, and economic activity. Like many other countries, the UK experienced severe operational difficulties in a number of industries, from supply chain disruptions to job volatility to forced company closures. This brought to light not just the weaknesses in different industries but also their ability to respond to significant unexpected shifts.
Long-term changes in business activities were sparked by the pandemic, which forced industries to modify their plans, embrace digital transformation, and implement new resilience techniques. Some industries struggled to continue operating, particularly those that rely largely on physical presence, while others easily shifted to remote labour and e-commerce. Given the complex nature and scope of the pandemic's effects, data-driven research is crucial for understanding the differences in sectoral impacts and guiding future risk awareness.
This project investigates the impact of the COVID-19 pandemic on five major sectors within the UK economy: **Accommodation and Food Services (Hospitality), Wholesale and Retail Trade (Retail), Manufacturing, Transportation and Storage, and Information and Communication**. It explores the effects across four critical dimensions: **financial performance**, **workforce dynamics**, **trade variations (import/export challenges)**, and **strategic business adaptations** during and after the height of the pandemic.

1.1 Problem statement

The COVID-19 pandemic brought an unexpected level of disruption that drastically changed the way businesses operate. Companies across the UK had to deal with sudden drops in revenue, workforce issues like furloughs, import and export delays, and an urgency to review their strategy in order to survive right away. Certain industries experienced greater damage than others, particularly those that depended on in-person services, but others were able to adjust more rapidly because of pre-existing digital infrastructure or flexible business models.

While there’s been a lot of public and government interest in how businesses were affected, much of the information available tends to focus on general trends. What's lacking is a deeper analysis of how particular industries reacted over time, not just during the pandemic's phases. This project fulfils that gap.

A useful source of information during the pandemic was the "Business Insights and Conditions Survey" (BICS), which was created by the UK's Office for National

Statistics (ONS). Businesses from a number of industries have contributed broad real-time feedback to this dataset, which offers insight into trade difficulties, workforce changes, and the usage of government initiatives. By examining this data, we have the opportunity to learn the true narrative of how industries handled the situation, both generally and by sector.

This project focuses on five key sectors that represent different types of business models and challenges: **Hospitality, Retail, Manufacturing, Transportation and Storage, and Information and Communication**. These sectors were chosen because they vary significantly in how dependent they are on physical presence, digital systems, and global trade, making them ideal for comparison.

This project is to highlight sector-specific challenges and recoveries throughout the use of exploratory data analysis, statistical testing, and developing of an interactive dashboard using Streamlit. Despite just being a technical exercise, this analysis provides valuable insights for future preparation, whether it be for researchers, policymakers, or business executives getting ready for the next unforeseen event. It also helps us understand what company resilience looks like under a global crisis.

1.2 Aims and Objectives

Aim:

The primary aim of this project is to investigate the long-term impact of the COVID-19 pandemic on the UK business landscape, using empirical data to evaluate how different sectors were affected and adapted over time. This study specifically focuses on financial performance, workforce dynamics, trade disruptions, and strategic responses, leveraging publicly available data from the **ONS** to provide sector-specific insights.

Objectives:

The aim of this project is to analyse important information that can guide plans for unforeseen circumstances in the future, advancing our knowledge of crisis management and corporate resilience.

- To examine how financial performance measured primarily through turnover change evolved during the pandemic for five key UK sectors.

- To analyse changes in workforce dynamics, with a focus on furlough rates and employment trends, and how businesses managed labour resources during pandemics.

- To evaluate the extent and nature of trade disruptions, especially in terms of import/export challenges, and how sectors mitigated supply chain issues.

- To identify and interpret the strategic responses businesses adopted (e.g., government schemes, remote working, digital transformation) in response to pandemic-related disruptions.

- To perform statistical analyses (e.g., t-tests) to assess significant changes between early and late pandemic phases, determining which sectors demonstrated resilience or vulnerability.

- To develop an interactive Streamlit dashboard that visually communicates the findings and allows stakeholders to explore sector-wise impacts and adaptations.

2. Background

2.1 Literature review

The world economy was significantly and, in many ways, impacted by the COVID-19 pandemic. Its impacts have been particularly disruptive for the UK in a number of commercial areas. Although government reports and academic literature have aimed to record the impacts, it has been challenging to develop a thorough knowledge of sector-specific trends, especially from a longitudinal data-driven perspective, due to the quickly changing environment throughout the pandemic.

Economic Disruption and Sector Vulnerability

Both the OECD (2021) and the World Bank (2020) highlighted how the pandemic affected consumer demand, disrupted global supply chains, and forced industries to shut down or drastically change their businesses. For instance, during the early 2020 lockdown, the hospitality industry experienced the worst recession, with UK GDP data indicating a 90%+ decline in activity (ONS, 2021). Due to customer uncertainty and mobility constraints, retail, especially non-essential retail, followed a similar pattern (Baker et al., 2020).

While industries like information and communication shown greater resilience due to digital infrastructure and remote work capabilities, manufacturing experienced delays because of dependency on foreign suppliers and a lack of workers (PwC, 2021). Border closures and shifts in logistics demand, especially with the rise in e-commerce, caused fluctuation in transportation and storage (McKinsey, 2020).

Workforce Challenges

The dynamics of the workforce also changed significantly. With more than 11.7 million jobs supported between March 2020 and September 2021, furlough measures under the UK Government's Coronavirus Job Retention Scheme (CJRS) were crucial in preventing a major unemployment crisis (HMRC, 2021). The scheme disproportionately benefited industries like hospitality and retail, where business continuity was impossible without human presence, according to academic studies like Blundell et al. (2021). This separated traditional economic indicators resulting in a unique situation where employment data stayed constant despite declining output.

Trade and Supply Chain Disruption

The impact of COVID-19 on global trade was the subject of several research. The idea of "supply-chain contagion," in which the failure of suppliers in one region

of the world spreads along value chains, is covered by Baldwin and Freeman (2020). The pandemic and post-Brexit regulatory issues caused further import delays in the UK, especially in industries like manufacturing and wholesale. A large number of businesses reported trade challenges with small and medium-sized businesses (SMEs) being the most at risk, according to the ONS Business Insights Survey.

Strategic Adaptation and Digital Transformation

During the pandemic, business strategies changed quickly. Digital adoption and remote work rules have been on the rise, especially in the information and communication industry, according to Brynjolfsson et al. (2020). In sectors like manufacturing and hospitality that depended on physical activities, this approach was less viable. Sector-specific differences also existed in the use of government assistance programs, such as tax breaks, loan guarantees, and business rate reductions (BEIS, 2021). Agile decision-making and the capacity to change strategies were essential for survival during uncertain times, according to studies like Kuckertz et al. (2020).

Use of Real-Time Business Data for Research

A key component of pandemic-related research has been the availability of real-time data from official statistical agencies. For presenting timely, specific industries insights into company operations throughout the pandemic, the ONS "Business Insights and Conditions Survey" (BICS) has received a lot of citations (ONS, 2022). This kind of real-time data, suggested by researchers like Xu et al. (2021), is crucial for both academic study and government, especially when it comes to analysing quickly changing crises.

However, few studies have carefully analysed this dataset to find statistically significant changes over time or to compare different sectors. The majority of current research either focusses on individual case studies or analyses macroeconomic indicators. Therefore, in order to capture the larger picture of pandemic impact and resilience, empirical, statistical analysis across different dimensions, such as turnover, workforce, trade, and strategy is required.

Table 1: Summary of Key Literature

Study / Report	Methods Used	Dataset	Key Insights
Blundell et al. (2021)	Empirical analysis	UK CJRS data	Furlough schemes softened employment shocks, heavily used in hospitality
Baldwin & Freeman (2020)	Theoretical modeling	Trade analysis	Supply chain fragility and global contagion effects
Brynjolfsson et al. (2020)	Survey-based study	U.S. business surveys	Rapid digital transformation in knowledge sectors
Kuckertz et al. (2020)	Qualitative interviews	Start-up responses	Strategic agility critical for business survival
ONS (2022)	Descriptive statistics	BICS data	Business impacts across turnover, workforce, and trade
BEIS (2021)	Government reporting	Policy adoption surveys	Sectoral use of government support schemes

Table 1: summary of Key Literature

2.2 Review of methods and applications

Researchers have used a variety of quantitative, statistical, and analytical techniques to understand the complex effects of a worldwide crisis such as COVID-19 on different business sectors. The type of data, the study's goals, and the sectoral scope all affect these approaches. This section examines important methods and their application in related research, evaluating their suitability for the project's focus on workforce dynamics, turnover, trade fluctuations, and strategic business adaptations across five key UK sectors.

Descriptive and Exploratory Data Analysis (EDA)

An essential first step in understanding data distributions, trends, and possible variations is exploratory data analysis. According to studies like Petropoulos and Makridakis (2020), EDA was applied throughout the pandemic in order to visualise patterns in employment, sales, and operational changes. In this research, sector-wise trends and seasonal changes in employment and turnover

between the early (2020) and late (2021) periods were identified using EDA techniques such as line charts, bar graphs, and summary statistics (mean, median, and standard deviation).
Given the faulty, real-world traits of the ONS BICS dataset, EDA also helps in feature selection by helping in the removal of variables with high missingness or low variance. Standard visual tools for this type of analysis, including Tableau dashboards or Python's Seaborn and Matplotlib, were used in this project to identify trends.

Hypothesis Testing and Statistical Significance
Several academic papers used t-tests and ANOVA to evaluate whether observed differences in variables before and after pandemic interventions were statistically significant. For example, Fernandes (2020) evaluated the impact of lockdown measures on sources of revenue in the retail and tourism industries across several nations using paired t-tests.
Key indicators, including turnover change, workforce furlough rates, and reported trade challenges, were compared between early and late pandemic periods in this study using independent two-sample t-tests. This method provided the empirical evidence of whether or not sectoral differences were statistically significant or simply observational. Verifying whether strategic or economic changes were a part of a larger trend, or just periodic fluctuations was particularly crucial.

Time-Series and Trend Analysis
Time-series analysis has frequently been used in related research to model the change of economic indicators over time, although this is not the primary goal of this research. Papers like those by Nicola et al. (2020) have forecasted economic recovery after COVID using tools including seasonal decomposition (STL), exponential smoothing, and ARIMA.
Since the goal of this research was comparative analysis rather than predicting, a more relatively easy type of time-series visualisation, line plots across time windows was selected. However, by grouping data chronologically and choosing split points for hypothesis testing according to time (e.g., early 2020 vs. mid/late 2021), the basic concept of time-aware analysis was protected.

Sector-Based Comparative Analysis
Comparing sectoral impacts to unexpected events is a popular approach in economics and policy research. For example, the OECD (2021) compared the responses of IT, hospitality, and manufacturing to workforce restrictions and supply chain disruptions.
By comparing five sectors (hospitality, retail, manufacturing, transportation, and information) across multiple crucial aspects, including financial performance, workforce dynamics, trade variation (import/export), and strategic adaptations,

the study thrived such an approach. It was feasible to figure out whether industries showed resilience against those that experienced chronic disturbance by aligning sectors and comparing metrics side by side.

Dashboarding and Interactive Visualisation

Storytelling dashboards, such as John Hopkins' COVID-19 tracker, gained popularity during the pandemic due to the increasing demand for real-time insights. Therefore, interactive data storytelling has been employed in a number of academic and governmental contexts using platforms like Tableau and Streamlit (e.g., UK COVID-19 Dashboard by Public Health England).

A multi-page dashboard featuring sectoral comparisons, early vs. late period visualisations, and an interactive data exploration module was created for this project using Streamlit. Dashboards were used as a communication tool to present data-driven insights to stakeholders that were not technical in addition to being a visual intervention.

Study / Source	Methodology Used	Application Scope	Relevance to This Project
Fernandes (2020)	T-tests and regression	Impact of COVID on EU economies	Validates the use of t-tests for policy-level differences
Nicola et al. (2020)	Time-series modeling and forecasting	Health and economy correlation	Reinforces the need for time-aware visualizations
OECD (2021)	Sectoral comparisons and dashboards	Global business disruptions	Useful for cross-sector alignment
Brynjolfsson et al. (2020)	Survey data and visual storytelling	Digital acceleration in firms	Influenced dashboard design choices
Petropoulos & Makridakis (2020)	Exploratory visual analysis	Retail sales data	Supported the use of EDA in this project

Table 2: Comparison of methods used in similar studies

Method	Strengths	Weaknesses
EDA & Visualisation	Quick insights, good for communication	Not statistically rigorous
T-tests	Simple and interpretable significance testing	Assumes normal distribution and independent samples
Time-series models	Forecasting capability, detects seasonality	Complex, requires large and consistent datasets
Sectoral comparison	Easy to communicate and interpret	Can overlook intra-sector variability
Interactive Dashboards	Engaging, facilitates exploration	Needs user knowledge to derive deep insight

Table 3: Strengths and Weaknesses of Methods

In conclusion, the techniques used in this project are EDA, t-tests, and sectoral comparisons presented via Streamlit to reflect an integration between accessibility and analytical accuracy. These techniques are appropriate for examining real-world ONS data, which may be non-linear, diverse, and dynamic, and they are in line with related studies.

2.3 Review of tools

A data-driven research project's successful execution depends on both the efficient use of technology tools and the selection of suitable methodology. To handle, analyse, and share massive information, professionals in data science, economics, and public policy have been using adaptable, open-source technologies more and more in recent years. An outline of the tools employed in this project is given in this part, along with an analysis of their functions in related research that looks at how disruptive incidents affect business activities.

Python (Pandas, Seaborn, Matplotlib, SciPy)

Python is a popular programming language for research and data analysis. It is a standard instrument in economic research because of its adaptability, wide library environment, and simplicity of integration. Python served as the main environment for statistical testing, exploratory analysis, and data preprocessing for this project.

- **Pandas** was essential for loading, cleaning, and transforming the ONS Business Insights and Conditions Survey (BICS) data, especially as it involved 25 different sector-specific files, later merged into five thematic datasets. Pandas supports time-series manipulation and multi-indexing, which were useful when comparing trends across sectors and dates.

- **Seaborn** and Matplotlib were used to generate high-quality visualizations, including line plots to show turnover trends, bar charts for furlough rates, and boxplots to visualize trade challenge distributions.
- **SciPy** enabled statistical hypothesis testing. Specifically, the independent two-sample t-tests used to compare early vs. late periods across key metrics relied on scipy.stats.ttest_ind.

These libraries have been employed in academic studies, who used similar Python-based workflows to analyse business behaviour during COVID-19. The reproducibility, flexibility, and scalability of Python made it a suitable choice for this study.

Streamlit

Streamlit is an open-source Python framework that transforms scripts **(Wrote in VS code)** into interactive web applications. In recent years, it has gained popularity for rapid dashboard prototyping, especially in academic and industry projects involving time-sensitive data such as COVID-19 tracking, sectoral recovery visualisation, or social policy response.

In this project, Streamlit was used to build a multi-page storytelling dashboard with the following features:

- **Introduction page**: Provides users with background context and research motivation.
- **Data visualisation page**: Presents pre-processed financial, trade, and workforce data in sectoral comparison plots.
- **Hypothesis testing insights**: Displays summary statistics and t-test results for early vs. late pandemic periods.
- **Interactive data exploration**: Allows users to select one of the five datasets, choose metrics, generate plots, and download insights.
- **Key takeaways and conclusion pages**: Summarise findings and implications in an accessible way.

Streamlit provides a Python-native interface that allows dynamic coding, real-time rendering, and direct integration of statistical tools, in contrast with Tableau or Power BI, which are often used in static or legal scenarios. A tool in line with modern distribution strategies, Streamlit has also been used by research teams like those at the OECD and university institutions for interactive visual storytelling throughout the pandemic.

Google Colab and GitHub

Google Colab served as the collaborative environment for all Python-based scripting. As a cloud-based platform, it provided GPU acceleration (where required), access to shared files, and Jupyter Notebook compatibility. Colab's integration with Google Drive ensured a backup of data cleaning and statistical analytics notebooks.

GitHub was used to do version-control the codebase and document project evolution. Academic and industry researchers alike use GitHub to collaborate on open-source COVID-related dashboards, simulation models, and data pipelines.

Its usage reflects good research practice by maintaining reproducibility and transparency in project development.

Tool	Role in this project	Strengths	Limitations
Python	Data wrangling, analysis, testing	Open-source, flexible, scalable	Learning curve for non-coders
Google Colab	Coding environment	Cloud-based, collaboration-friendly	Requires internet, slower for very large files
VS Code	Dashboard code	Efficient for data-driven project	Setting up the env & dependencies are tricky
Streamlit	Dashboard creation and storytelling	Interactive, Python-native, deployable	Less advanced than dedicated BI tools
GitHub	Code versioning and documentation	Enables transparency and collaboration	Public by default unless private repo used

Table 4: Comparison of Tools

3. Legal, Ethical, Sustainability and other considerations

A wide range of legal, ethical, and professional issues must be taken into consideration when developing and delivering a data science project that uses sensitive or socioeconomically significant data. This section describes how these factors were incorporated into the current study, which uses official statistics from the UK's Office for National Statistics (ONS) to assess the COVID-19 pandemic's sectoral business effects.

3.1 Legal Considerations

The **ONS's Business Insights and Conditions Survey (BICS)** served as the project's main source of data. **The Open Government License v3.0**, which allows usage, modification, and distribution with proper credit, explicitly states that the ONS's data is publicly accessible. This project entirely corresponds with that license: no attempts were made to commercialise, modify, or claim ownership of the underlying datasets, and all data was properly cited.

Additionally, there were no concerns regarding **personally identifiable information (PII)** because the data was anonymised and aggregated at the industry level, which made the project compliant with data protection laws like the **Data Protection Act of 2018** and the **UK General Data Protection Regulation (UK GDPR)**.

Furthermore, **no third-party or proprietary datasets were introduced** into the workflow, ensuring the analysis remained legally sound and open for reproducibility and peer review.

All journal results, scripts, and dashboards were stored in publicly available repositories with complete licensing disclaimers to further improve legal compliance. This ensures the traceability of all analytical procedures in addition to their legality.

3.2 Ethical Considerations

Beyond legislation, ethics in data science involves transparency, accountability, and equality. Several ethical factors were considered in this undertaking, including:

- **Transparency**: To guarantee transparency, GitHub was used to version-control and document all code, datasets, and visualisations. End users can examine insights in an interpretable and visually engaging way with the finished Streamlit dashboard.

- **Avoiding misinterpretation**: Care was taken while phrasing insights because the data relates to delicate economic fluctuations. To prevent excessively positive or alarmist readings, descriptive text was used together with visualisations. Informing, not affecting, was the goal.

- **Responsibility in Communication**: Social and emotional weight is attached to strategic business adjustments and workforce changes, such as furloughing. To prevent biassed or disrespectful stories, definitions was carefully chosen throughout the dashboard and written analysis.

- **No harm principle**: No experimental or manipulative procedures were conducted on human subjects. All findings reflect secondary data analysis.

- **Bias Awareness**: Since algorithmic bias may arise even in descriptive analytics, care was taken to treat all industries equally and prevent from praising or criticising any one of them based on inadequate data.

- **Inclusion and Equity**: Regardless of technical expertise, a broad range of audiences, including students, policymakers, and other stakeholders, could use the dashboard thanks to the use of open tools and accessible design choices.

3.3 Sustainability Considerations

In data science, sustainability can be seen from a social and environmental aspect. Wider digital sustainability criteria were somehow followed even if this project's environmental impact was naturally minimal because it made use of cloud platforms and publicly accessible data:

- **Minimal computational waste**: Lightweight computing was ensured by the use of Python, Streamlit, and Google Colab. By avoiding heavy models or cloud services, unwanted resource usage was reduced.

- **Open Science principles** Through community involvement and knowledge reuse, the project complies with sustainability standards by applying open-source tools and releasing all resources on GitHub.

- **Long-term usability**: The dashboard is a reusable tool for researchers, students, or policymakers looking at post-pandemic business recovery because it was made compatible with future updates or data integrations.

- **Digital sustainability**: Because Colab and Streamlit are widely used in both professional and educational contexts, and because they support a wide range of Python versions, the tools selected ensure the project can be carried out even as platforms change.

- **Educational reusability**: To ensure that the knowledge created continues to help others, the report and dashboard can also be used as educational resources by students or analysts in their early careers who are researching post-pandemic recovery.

3.4 Social and Professional Responsibility

As a student of data science and analytics, adherence to professional ethical standards was an integral part of the project design. Among them were:

- **Respect for academic integrity**: The Harvard reference style was used to cite all sources, whether they were data repositories, internet documentation, or scholarly literature.

- **Reproducibility and peer review readiness**: The project was created with a clear framework, and analytical procedures were recorded to enable peer replication.

- **Accessibility**: Readability and usability were key considerations in the dashboard's design. To guarantee accessibility for stakeholders who are not technical, visuals employ sector colour coding, consistent labelling, and simple language.

- **Awareness of unintended consequences**: The dashboard shows trends rather than suggestions, leaving it up to knowledgeable users to interpret. This stops data-driven conclusions from being abused for biassed narratives or unsupportable policymaking.

- **Professional Presentation**: In addition to being a technical solution, its outcome was presented as a polished, thoroughly documented application that could be used to demonstrate academic and professional skills.

So, ethical reflection was not treated as a checklist item, but as an ongoing consideration across every project phase from data selection to dashboard delivery.

4. Methodology

This project followed a structured data science methodology designed to investigate how five major UK business sectors: **Hospitality**, **Retail**, **Manufacturing**, **Transportation and Storage**, and **Information and Communication**. The COVID-19 pandemic had an impact on them. Using publicly accessible real-time company data, the research attempted to investigate shifts in workforce dynamics, trade challenges, financial performance, and strategy responses. Data collection, cleansing, standardising, exploratory data analysis (EDA), hypothesis testing, and creating a storytelling dashboard were all done in compliance with the typical data science lifecycle.

Python was used as the main programming language for the project, and it was developed using Visual Studio Code (VS Code) and Google Colab. Colab was used for code execution, visualisation, and exploratory work, while Visual Studio Code was used for dashboard development and integration with the Streamlit framework.

4.1 Data

Data Sources and Scope

The Business Insights and Conditions Survey (BICS) of the Office for National Statistics (ONS), an extensive database of sector-specific indicators illustrating the pandemic's effects on UK businesses, provided the dataset that was used. From 2020 onwards, the survey provided monthly and weekly updates on government support utilisation, employment, trade challenges, and turnover.

The project focused on five main categories:

- **Financial Performance (Turnover)**= July 2020 to December 2021
- **Workforce Dynamics (Furlough Use)**= July 2020 to June 2021
- **Trade Variations (Import and Export Challenges)** = October 2020 to June 2021
- **Strategic Adaptations (Government Schemes and Business Decisions)**= July 2020 to September 2021

Why These Sectors Were Chosen

The five sectors selected for this study were:

- Accommodation and Food Service Activities (Hospitality)
- Wholesale and Retail Trade (Retail)
- Manufacturing

- Transportation and Storage
- Information and Communication

These industries were specifically picked because they reflect a wide range of the UK economy, with significant variances in their operational processes and levels of pandemic exposure:

- **Hospitality** was one of the first industries to be forced to close because of social distance regulations, which mostly depends on direct connection. It acts as a standard for industries with a lot of contact.

- **Retail** had experienced an equivalent lockdown impact but had more opportunity to switch to online sales. It provides a comparison between traditional and digital commerce.

- **Manufacturing** was more flexible in putting on-site procedures and some automation in place but also faced operational and supply chain challenges.

- **Transportation and Storage** was essential during the pandemic due to the expansion in logistics and delivery services, yet it was also restricted by border closures and fuel shortages.

- **Information and Communication** stood out as the most digitally robust sector, with broad usage of remote working and no interruption to service delivery.

The study was able to investigate a broad range of sectoral behaviours and resilience levels during the pandemic by selecting industries with differing degrees of physical dependency, digital infrastructure, and exposure to international trade.

Why These Themes Were Chosen?

The study focused on four key themes:

1. Financial Performance (Turnover)

2. Workforce Dynamics (Furlough Use)

3. Trade Variations (Import and Export Challenges)

4. Strategic Adaptations (Use of Government Schemes and Business Decisions)

These themes were selected because they reflect **core areas of business health and survival** during a crisis:

- **Turnover** is a clear indicator of both operational continuity and financial performance. By examining turnover, one may determine which industries recovered more quickly and which still faced difficulties.

- **Furlough** usage shows that why The CJRS is a clear sign of workforce changes. It assists in measuring labour market shocks and assessing the success of government initiatives in preserving jobs.

- **Supply chain** vulnerabilities and the ways in which global trade dependencies influenced sector performance were revealed by import/export challenges.

- **Strategic adaptations** provided clarity on how companies handled inconsistencies, whether it was by reorganising their services, embracing digital tools, moving operations, or using government help.

Why These Dates Were Chosen?
Each theme category's specific time frames were selected based on the availability of data, their relevance to the pandemic timeline, and their compatibility with significant stages of business and governmental reaction:

- **Financial Performance (Turnover): July 2020 to December 2021**
  The post-lockdown recovery and the increase of the Delta variation are just two examples of the many economic changes that occurred during this time. It was chosen in order to track the evolution of turnover through various reopening phases, restrictions, and adjustments to policies such as extended furlough and Eat Out to Help Out.

- **Workforce Dynamics (Furlough Use): July 2020 to June 2021**
  Because they capture the crucial period of the **Coronavirus Job Retention Scheme (CJRS)** prior to it decreasing these dates were selected. Furlough usage sharply decreased by July 2021, by which time many companies had either permanently changed workforce levels or returned to regular operations.

- **Trade Variations (Import and Export Challenges): October 2020 to June 2021**
  Trade metrics were tracked from October 2020 due to increased reporting by businesses on import/export challenges This aligns with the transitional changes caused by Brexit, which started to affect trade

dynamics in combination with pandemic disruptions. A constant cut-off before the data structure changed was June 2021.

- **Strategic Adaptations (Government Schemes and Business Decisions): July 2020 to September 2021**
  This range contains the critical time frame during which companies actively used government support programs (such as Kickstart and Job Retention Bonus) and started modifying their plans in reaction to ongoing uncertainty. As many schemes were being phased out or transformed by September 2021, comparative study naturally came to an end at that point.

Individual datasets for the five sectors under study were included in each subject. These CSV files served as the most important input for the analysis and were taken straight from ONS releases.

Variables and Format…
All datasets had the same structure:

- A **date column** (monthly or every two weeks)
- A **sector/category column**
- One or more based on percentage, such as:
  - % of businesses reporting a decrease in turnover
  - % of workforce on furlough
  - % of businesses dealing with export or import disruptions
  - % of businesses using particular government schemes

Data Cleaning and Merging…
Pandas and NumPy were used in Python to clean and adjust the data. The steps that were taken:

- Standardising column names across all CSVs
- Converting dates to datetime format (inconsistent formats)
- Forward or backward filling missing values
- Dropping unnecessary or redundant columns
- Ensuring percentage values were within acceptable ranges (0–100%)
- Reordering and sorting datasets chronologically

Once cleaned, the 25 datasets were **merged** by category across sectors to create five consolidated files: (Merged_Data)

- Merged_Financial.csv
- Merged_Import.csv
- Merged_Export.csv
- Merged_Workforce.csv
- Merged_Strategic.csv

These merged datasets enabled cross-sectoral comparison on similar metrics and were used throughout the project's analysis and dashboard.

4.2 Methods

Exploratory Data Analysis (EDA)
EDA was the initial phase of the analytical process It aimed to identify trends, patterns, and anomalies in the data that were specific to the sector. The methods and visualisations listed below were used:

- **Line plots** used to track changes in metrics like turnover and furlough over time
- **Combined line plots** used to discover how main factors got affected
- **Descriptive statistics** (mean, median, standard deviation) for summarising each variable

Key early insights from EDA included:

- A **steep drop in turnover** in the hospitality and retail sectors during lockdown periods
- **Furlough usage** peaking in transportation and hospitality
- **Trade disruptions** affecting manufacturing and retail the most

These trends were visualised using Python libraries such as **Matplotlib** and **Seaborn** and later included as static and interactive charts in the dashboard.

Hypothesis Testing (Early vs. Late Comparison)
In order to statistically validate sectoral differences over time, **independent two-sample t-tests** were applied. Each metric was compared across two defined periods:

- **Early Period**: July 2020 – December 2020
- **Late Period**: January 2021 – post-June 2021 (depending on data availability)

The goal was to check whether differences observed in the data were statistically significant. The **SciPy.stats** module was used for this purpose, with a **significance level (α) of 0.05**.

Findings:

- **Significant reduction in furlough rates** post-2021 for all sectors
- **Improved turnover** in hospitality and retail with reopening
- **Minimal change** in export challenges across most sectors, indicating persistent issues

All t-test results were captured, explained, and visualised in a dedicated dashboard page.

Algorithm and Function Development

A number of Python functions were developed in order to ensure reusability and automate analysis:

- **Data merging functions** to create merged datasets from sectoral CSVs
- **Plot generation features** for sector-specific charts
- **Statistical comparison functions** to run and summarise t-test findings across sectors

These features were modular and made to be used in the Streamlit app as well as in notebooks. Before the full-scale implementation, sample data was used for testing and validation.

Streamlit Dashboard Development

Using Streamlit to create an interactive, multi-page dashboard was one of the project's key results. Visual Studio Code was used to create and test the dashboard, using reusable components and Python scripts.

Features of the Dashboard:

- **Introduction Page**: Explained project background and objectives
- **Sector Insights Page**: Displayed visual comparisons of each sector's metrics
- **Hypothesis Testing Page**: Showed early vs. late period results with statistical interpretation
- **Key Takeaways Page**: Allowed users to view summarised insights by sector
- **Interactive Exploration Page**: Enabled users to load any of the five merged datasets, select metrics, view charts, and download data or visuals
- **Conclusion Page**: Provided a final reflection and summary of the findings

The dashboard enabled a ***storytelling approach*** to data, useful for both technical and non-technical stakeholders. Built-in download buttons allowed

users to extract CSV data from the sectors and themes they chose and save interactive Plotly charts.

Testing and Deployment

The dashboard was tested across a number of browsers and devices. To ensure usability, every feature from generating charts to downloading data was examined. Iterative adjustments were made for issues such mismatched columns, empty plots, and incorrect metric choices.

Despite being developed locally, the dashboard's framework permits future deployment on cloud platforms like Heroku or Streamlit Cloud with very few modifications.

Project Workflow and Tools Summary

- **Google Colab** was used for initial data cleaning, EDA, and hypothesis testing
- **Visual Studio Code** was used for writing, testing, and managing Streamlit application scripts
- **GitHub** was used for version control and sharing the complete project (notebook files, dashboard scripts, and outputs)
- **Streamlit** enabled the development of a fully interactive dashboard

All analysis scripts and visual components were reproducible and documented

Conclusion of Methodology

This project's analytical approach was comprehensive, maintaining a balance between visual storytelling and statistical rigour. Each step, from real-time government data to hypothesis-driven analysis and an interactive dashboard accessible to the public, helped to clarify the various ways in which the pandemic impacted various industries. The insights produced are made accessible and practical through the use of clear, modular Python code and interactive tools, providing a basis for further study and policymaking.

5. Tools

A combination of modern, open-source tools and platforms were used in this project to perform data analysis, hypothesis testing, and the development of an interactive dashboard. These tools were selected based on their efficiency, accessibility, scalability, and relevance to the tasks at hand. This section outlines the tools used and the rationale behind their selection.

Python Programming Language

Python served as the core environment for analysis, as discussed in the Methodology section. Key libraries such as Pandas (for data handling), Seaborn and Matplotlib (for visualisation), and SciPy (for t-tests) were used:

Pandas was the primary tool for handling the datasets. It provided efficient data structures for manipulating large CSV files, allowing operations such as filtering, standardising, grouping, merging, and time-series analysis. Pandas was also instrumental in cleaning and formatting the data extracted from the Office for National Statistics (ONS) Business Insights and Conditions Survey (BICS).

- **NumPy** was used in support of Pandas for numerical computations. It allowed for efficient handling of arrays and mathematical operations, which were crucial during the preprocessing stage and when calculating statistical metrics.

- **Matplotlib and Seaborn** were used to create detailed visualisations. While Matplotlib enabled the production of customisable plots and charts, Seaborn provided high-level functionality for statistical visualisation, making it easier to identify patterns, trends, and outliers across sectors.

- **SciPy**, particularly the scipy.stats module, was used to conduct independent two-sample t-tests. These tests formed the basis of the hypothesis testing portion of the project, where early and late periods were compared to identify significant differences in business metrics across sectors.

Google Colab for Development and Collaboration

Google Colab served as the cloud-based development environment. It offered a browser-based interface similar to Jupyter Notebook, with the added benefits of automatic saving, easy sharing, and access to free computational resources. This environment enabled the execution of all data cleaning, statistical analysis, and visualisation code, and was particularly helpful during the exploratory data analysis phase. It also supported the inclusion of markdown cells, which allowed documentation alongside code for clarity and reproducibility.

Visual Studio Code (VS Code)

All Streamlit application code was written and iteratively tested in **Visual Studio Code**, a popular and lightweight integrated development environment (IDE). VS Code was chosen for its powerful code editing features, real-time linting, version control integration (with GitHub), and rich extension marketplace. Its ability to manage Python virtual environments and detect syntax issues helped streamline development. Furthermore, the terminal integration allowed for seamless testing of the dashboard using local Streamlit servers.

Streamlit for Dashboard Development

Streamlit was used to transform the analysis into an interactive, multi-page dashboard. Rapid development was made possible using Streamlit, a Python-based web application framework that eliminated the need for traditional front-end development. Users were given a guided tour of the project findings via the dashboard. Interactive data exploration, downloadable graphics, and sector-by-sector comparisons were among the features. Streamlit is a perfect tool for sharing insights because of its lightweight architecture, which allowed the application to run locally or be deployed online with little setup.

Justification for Tool Selection

Each tool was chosen strategically based on its strengths and suitability for the project requirements:

- **Python** provided the analytical backbone, known for its simplicity, readability, and powerful libraries, making it ideal for processing large datasets and conducting complex statistical analysis.
- **Google Colab** enabled real-time collaboration, making it easy to test and iterate on code across devices without dependency issues or installation requirements.
- **VS Code** ensured a stable and efficient development environment, especially when building and debugging the dashboard.
- **Streamlit** offered a fast, intuitive way to share insights interactively. It supported the creation of a dashboard that could be used by a wide audience without requiring technical expertise.

Together, these tools contributed to a workflow that was efficient, reproducible, and well-suited to both technical analysis and presentation, ensuring the successful execution of the project's goals.

6. Model development

A complete overview of the model development procedure used in the project is given in this section. In this context, the term "model" refers to the broader analytical infrastructure developed to enable descriptive analysis, statistical testing, and interactive reporting of business impacts during the COVID-19 pandemic, even though it typically refers to predictive or machine learning algorithms.

The model development involved three major phases:
1. “Data Processing Pipeline”
2. “Statistical Hypothesis Testing”
3. “Streamlit Dashboard Development”

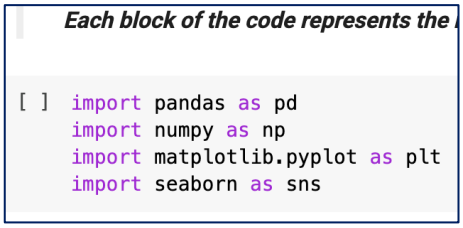
Each phase is described in detail below, along with the techniques, tools, and logic applied throughout the process…

6.1 Data Processing Pipeline

The preparation pipeline is the foundation of any successful data project. The pipeline was created to automatically import, clean, normalise, and merge datasets across five sectors and four themes due to the Office for National Statistics' (ONS) huge amount of sector-specific data. Establishing a logical, analytically solid framework that could accommodate both longitudinal and cross-sectional findings was the aim.

6.1.1 Data Import and Cleaning : For scripts, refer to EDA.ipynb

Building on the cleaning process described in Methodology, multiple sector-specific CSV files from ONS BICS were standardised, merged, and prepared for model development using Pandas.



Important libraries were called at first…

Figure 1: Imported libraries in EDA

Key steps included:

1. **Importing Files:** Each CSV was loaded and previewed using pd.read_csv() and .head() to understand the structure.



Figure 2: Importing files in EDA

2. **Converting Date Columns:** All date fields, originally strings, were converted into datetime objects via pd.to_datetime() to support time-series visualisation and filtering. All date columns were aligned to a consistent frequency (monthly or fortnightly, depending on the original format).

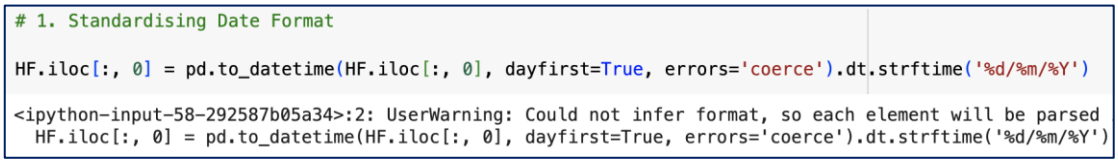


Figure 3: Script to standardise date format

Dates before standardising

	Date
0	2/7/20
1	16/07/2020
2	30/07/2020
3	13/08/2020
4	27/08/2020
5	10/9/20
6	24/09/2020

Dates after standardising

	Date
0	02/07/2020
1	16/07/2020
2	30/07/2020
3	13/08/2020
4	27/08/2020
5	10/09/2020
6	24/09/2020

3. **Standardising '*' with NaN:** Missing values were replaced using .replace() to create uniformity across datasets.

```
# 2. Replacing '*' with NaN

HF.replace('*', np.nan, inplace=True)
```

Figure 4: Script to replace '' with NaN*

4. **Standardising column names by adding '_' at space:** To maintain the format of column names.

```
# 3. Replacing spaces with underscores in column names

HF.columns = HF.columns.str.replace(" ", "_")
```

```
['Date',
 'Turnover_Has_Increased_By_More_Than_50%',
 'Turnover_Has_Increased_Between_20%_And_50%',
 'Turnover_Has_Increased_By_Up_To_20%',
 'Turnover_Unaffected',
 'Turnover_Has_Decreased_By_Up_To_20%',
 'Turnover_Has_Decreased_Between_20%_And_50%',
 'Turnover_Has_Decreased_By_More_Than_50%',
 'Not_Sure_']
```

Figure 5: Script to standardise column names

5. **Normalising Percentages:** Several metrics were in percentage format and were normalised to fall between 0 and 100, ensuring comparability across datasets.

```
[ ] # 4. Converting percentage columns to float

    cols_to_convert = HF.columns[1:]  # Have to exclude 'Date' column
    HF[cols_to_convert] = HF[cols_to_convert].replace('%', '', regex=True).astype(float)
```

```
HF.dtypes
```

	0
Date	object
Turnover_Has_Increased_By_More_Than_50%	float64
Turnover_Has_Increased_Between_20%_And_50%	float64
Turnover_Has_Increased_By_Up_To_20%	float64
Turnover_Unaffected	float64
Turnover_Has_Decreased_By_Up_To_20%	float64
Turnover_Has_Decreased_Between_20%_And_50%	float64
Turnover_Has_Decreased_By_More_Than_50%	float64
Not_Sure_	float64

dtype: object

Figure 6: Script to normalise percentages

6. **Handling Missing Values:** Missing values were addressed using **forward-fill and backward-fill methods** (.ffill() and .bfill()), as these techniques preserved the natural continuity of time-series data without introducing artificial spikes, assumptions, or biases that could distort trend analysis.

```
HF.ffill(inplace=True)  # First, try forward fill
HF.bfill(inplace=True)  # Then, backward fill as a backup
HF.interpolate(method='linear', inplace=True)  # Finally, interpolate remaining gaps
```

```
<ipython-input-77-530a0d42b6f1>:3: FutureWarning: DataFrame.interpolate with object dty
  HF.interpolate(method='linear', inplace=True)  # Finally, interpolate remaining gaps
```

Figure 7: Script to handle missing values

6.1.2 Plotting the Data to understand trends over time.

```
# Set 'Date' as index...
HF.set_index("Date", inplace=True)

# Plotting the trends...
plt.figure(figsize=(12, 6))
for col in HF.columns:
    plt.plot(HF.index, HF[col], label=col)

plt.xlabel("Date")
plt.ylabel("Turnover Percentage")
plt.title("Turnover Trends Over Time")
plt.xticks(rotation=45)
plt.grid(True)
plt.legend()
plt.show()
```

Figure 8: Script to plot trends

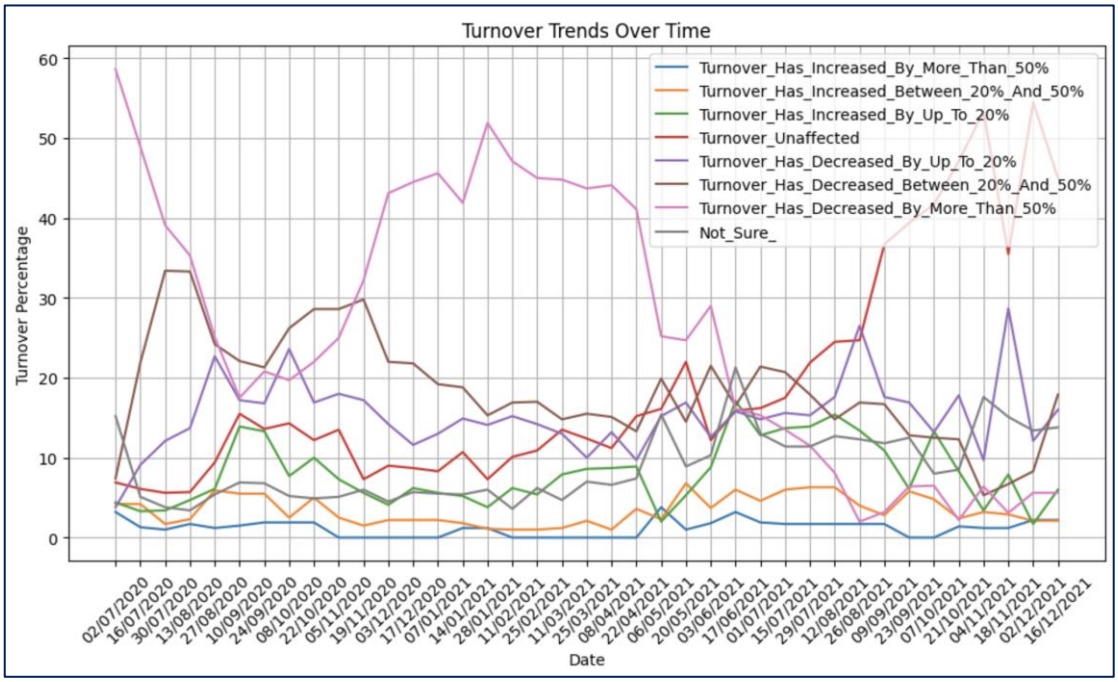


Figure 9: Turnover trends over time

Which helped understand how each sector behaved during COVID-19.

Now plotting grouped trends over time to visualise major columns…

```
HF["Turnover_Increased"] = HF.iloc[:, 0:3].sum(axis=1)
HF["Turnover_Decreased"] = HF.iloc[:, 4:7].sum(axis=1)

plt.figure(figsize=(12, 6))
plt.plot(HF.index, HF["Turnover_Increased"], label="Increased Turnover", color="green")
plt.plot(HF.index, HF["Turnover_Unaffected"], label="Unaffected Turnover", color="blue")
plt.plot(HF.index, HF["Turnover_Decreased"], label="Decreased Turnover", color="red")

plt.xlabel("Date")
plt.ylabel("Percentage")
plt.title("Grouped Turnover Trends")
plt.xticks(rotation=45)
plt.legend()
plt.grid()
plt.show()
```

Figure 10: Script to create grouped trends in turnover

These grouped trends helped understand if the particular theme were increased/decreased or stayed unaffected. As below we can see that the turnover was increased but later started decreasing in the middle of pandemic period, later ended up staying unaffected.

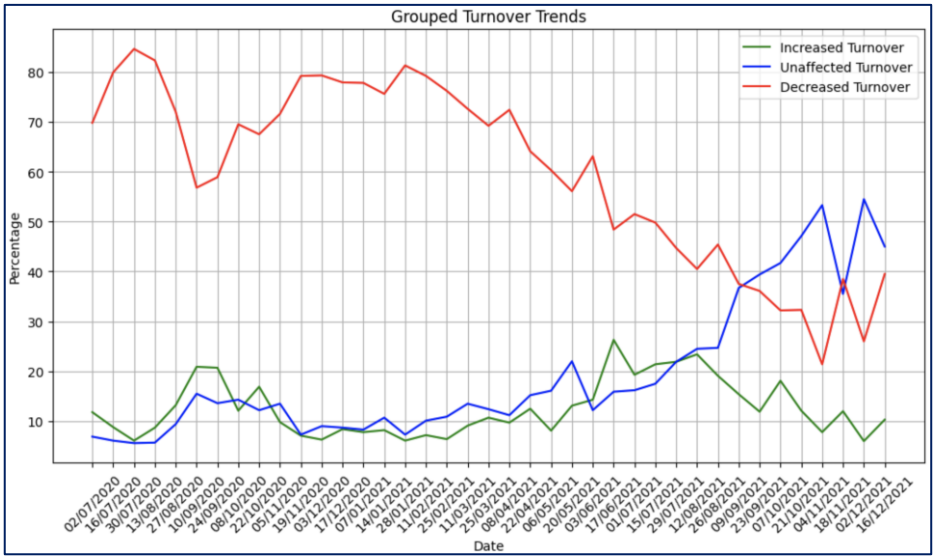


Figure 11: Grouped turnover trends

Also compared whether turnover dropped more in early pandemic or late pandemic?

```
HF["Period"] = ["Early Pandemic" if date <= "2021-03-01" else "Post-Pandemic" for date in HF.index]
plt.figure(figsize=(12, 6))
sns.boxplot(x="Period", y="Turnover_Decreased", data=HF)
plt.title("Comparison of Business Turnover Drop Before and After March 2021")
plt.show()
```

Figure 22: Script to compare early or post pandemic turnover

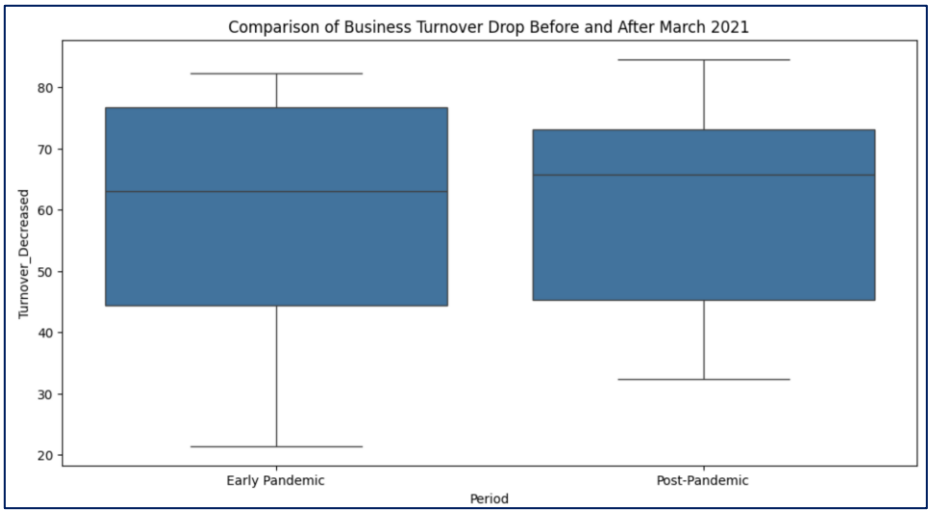


Figure 33: Boxplot of comparing business turnover drop.

Checked when turnover faced peak decline and incline…

```
print("Peak turnover decline:", HF["Turnover_Decreased"].idxmax())
print("Peak turnover recovery:", HF["Turnover_Increased"].idxmax())
```

```
Peak turnover decline: 30/07/2020
Peak turnover recovery: 17/06/2021
```

Figure 14: script to check peak decline & incline in turnover

EDA maintained the similar model throughout all sectors and theme of the project. For the rest of the project, this preprocessing stage produced a solid and consistent data foundation.

Themes:

Export:

```
# Summing up relevant columns for grouped insights
IE["Export_Decreased"] = IE["The_business_has_stopped_exporting_in_the_last_two_weeks"] + IE["The_business_is_exporting,_but_less_than_normal"]
IE["Export_Unchanged"] = IE['Exporting_has_not_been_affected']
IE["Export_Increased"] = IE['The_business_is_exporting,_but_more_than_normal']

# Plotting grouped import trends
plt.figure(figsize=(12,6))
plt.plot(IE.index, IE["Export_Decreased"], label="Decreased Export", color="red")
plt.plot(IE.index, IE["Export_Unchanged"], label="Unchanged Export", color="blue")
plt.plot(IE.index, IE["Export_Increased"], label="Increased Export", color="green")

plt.xlabel("Date")
plt.ylabel("Export Percentage")
plt.title("Grouped Information Export Trends")
plt.legend()
plt.xticks(rotation=45)
plt.grid()
plt.show()
```

Figure 45: Script to create grouped trend in export

Import:

```
# Summing up relevant columns for grouped insights...
HI["Import_Decreased"] = HI["Not_been_able_to_import_in_the_last_two_weeks"] + HI["Importing,_but_less_than_normal"]
HI["Import_Unchanged"] = HI["Importing_has_not_been_affected"]
HI["Import_Increased"] = HI["Importing_more_than_normal"]

# Plotting grouped import trends...
plt.figure(figsize=(12,6))
plt.plot(HI.index, HI["Import_Decreased"], label="Decreased Imports", color="red")
plt.plot(HI.index, HI["Import_Unchanged"], label="Unchanged Imports", color="blue")
plt.plot(HI.index, HI["Import_Increased"], label="Increased Imports", color="green")

plt.xlabel("Date")
plt.ylabel("Import Percentage")
plt.title("Grouped Hospitality Import Trends")
plt.xticks(rotation=45)
plt.legend()
plt.grid()
plt.show()
```

Figure 56: Script to create grouped trends in import

Workforce:

```
# Creating grouped categories...
HW["Not_Working"] = HW["On_furlough_leave_"] + HW["Off_sick_or_in_self-isolation_due_to_coronavirus_(COVID-19)_with_statutory_or_company_pay_"]
HW["Working_Normally"] = HW["Working_at_their_normal_place_of_work_"]
HW["Working_Remotely"] = HW["Working_remotely_instead_of_at_their_normal_place_of_work_"]

# Plotting workforce distribution trends...
plt.figure(figsize=(12,6))
plt.plot(HW.index, HW["Not_Working"], label="Not Working", color="red")
plt.plot(HW.index, HW["Working_Normally"], label="Working Normally", color="blue")
plt.plot(HW.index, HW["Working_Remotely"], label="Working Remotely", color="green")

plt.xlabel("Date")
plt.ylabel("Workforce Percentage")
plt.title("Workforce Status Trends in Hospitality")
plt.legend()
plt.xticks(rotation=45)
plt.grid()
plt.show()
```

Figure 67: Script to create grouped trends in workforce

Strategic:

```
# Summing up relevant columns for grouped insights...
HS["Using_Schemes"] = HS["Job_Retention_Bonus"] + HS["Kickstart_Job_Scheme_for_young_people"]
HS["Not_Sure"] = HS["Not_sure_"]
HS["Not_Using_Schemes"] = HS["Not_using,_or_intending_to_use,_either_of_these_schemes"]

# Plotting grouped import trends...
plt.figure(figsize=(12,6))
plt.plot(HS.index, HS["Using_Schemes"], label="Schemes Using", color="red")
plt.plot(HS.index, HS["Not_Sure"], label="Not Sure", color="blue")
plt.plot(HS.index, HS["Not_Using_Schemes"], label="Not Using Any Schemes", color="green")

plt.xlabel("Date")
plt.ylabel("Strategic use Percentage")
plt.title("Grouped Hospitality Strategic Trends")
plt.xticks(rotation=45)
plt.legend()
plt.grid()
plt.show()
```

Figure 78: Script to create grouped trends in strategic

6.2 Statistical Testing & Hypothesis Validation: For scripts, refer to hypothesis.ipynb

6.2.1 Dataset Merging

To enable comparative, cross-sectoral analysis, it was necessary to combine sector-specific datasets into unified theme-based datasets. For example, all sector-wise turnover data were merged into a single Merged_Financial.csv file. Unlike a simple concatenation of rows, this project required a column-wise merge, aligning sectors by date to facilitate time-based visual comparisons.

```
# "Financial"

# Defining file paths and sector names
sector_files = {
    'Hospitality': '/content/drive/MyDrive/Cleaned_FYP/HF_cleaned.csv',
    'Retail': '/content/drive/MyDrive/Cleaned_FYP/RF_cleaned.csv',
    'Manufacturing': '/content/drive/MyDrive/Cleaned_FYP/MF_cleaned.csv',
    'Transport': '/content/drive/MyDrive/Cleaned_FYP/TF_cleaned.csv',
    'Information': '/content/drive/MyDrive/Cleaned_FYP/IF_cleaned.csv'
}

dfs_finance = []

for sector, path in sector_files.items():
    df = pd.read_csv(path)

    # Renaming all columns except 'Date'
    df = df.rename(columns={col: f"{sector}_{col}" for col in df.columns if col != 'Date'})

    dfs_finance.append(df)

# Merging all dataframes on 'Date'
df_financial = reduce(lambda left, right: pd.merge(left, right, on='Date', how='outer'), dfs_finance)
```

Figure 89: Script to merge sectoral files on the basis of theme.

The merging logic was implemented in Python using the following approach:

- A dictionary (sector_files) was created, mapping each sector name (e.g., Hospitality, Retail) to its corresponding cleaned CSV file path.
- Each CSV was loaded into a Pandas dataframe, and a unique prefix (based on the sector name) was added to each column except the 'Date' column. This ensured there were no column name conflicts after merging and that each column could be clearly traced back to its sector.
- All dataframes were stored in a list (dfs_finance) and then merged on the common Date column using functools.reduce() and pd.merge() in an outer join format. This preserved all date entries from all sectors, even if some were missing in others.
    - Merged_Financial.csv
    - Merged_Workforce.csv

- Merged_Import.csv
- Merged_Export.csv
- Merged_Strategic.csv

These consolidated files were saved for use in analysis scripts and the Streamlit dashboard.

- It maintained **column-level traceability**, making post-merge analysis clear and manageable.
- Using **outer joins** ensured no loss of information from any sector's timeline.
- The resulting dataframe enabled **multi-sector visualisation** on the same time axis without the need for repeated manual alignment.

```
df_financial.columns

Index(['Date', 'Hospitality_Turnover_Has_Increased_By_More_Than_50%',
       'Hospitality_Turnover_Has_Increased_Between_20%_And_50%',
       'Hospitality_Turnover_Has_Increased_By_Up_To_20%',
       'Hospitality_Turnover_Unaffected',
       'Hospitality_Turnover_Has_Decreased_By_Up_To_20%',
       'Hospitality_Turnover_Has_Decreased_Between_20%_And_50%',
       'Hospitality_Turnover_Has_Decreased_By_More_Than_50%',
       'Hospitality_Not_Sure_',
       'Retail_Turnover_has_increased_by_more_than_50%',
       'Retail_Turnover_has_increased_between_20%_and_50%',
       'Retail_Turnover_has_increased_by_up_to_20%',
       'Retail_Turnover_unaffected',
       'Retail_Turnover_has_decreased_by_up_to_20%',
       'Retail_Turnover_has_decreased_between_20%_and_50%',
       'Retail_Turnover_has_decreased_by_more_than_50%', 'Retail_Not_sure_',
       'Manufacturing_Turnover_has_increased_by_more_than_50%',
       'Manufacturing_Turnover_has_increased_between_20%_and_50%',
       'Manufacturing_Turnover_has_increased_by_up_to_20%',
       'Manufacturing_Turnover_unaffected',
       'Manufacturing_Turnover_has_decreased_by_up_to_20%',
       'Manufacturing_Turnover_has_decreased_between_20%_and_50%',
       'Manufacturing_Turnover_has_decreased_by_more_than_50%',
       'Manufacturing_Not_sure ',
       'Transport_Turnover_has_increased_by_more_than_50%',
       'Transport_Turnover_has_increased_between_20%_and_50%',
       'Transport_Turnover_has_increased_by_up_to_20%',
       'Transport_Turnover_unaffected',
       'Transport_Turnover_has_decreased_by_up_to_20%',
       'Transport_Turnover_has_decreased_between_20%_and_50%',
       'Transport_Turnover_has_decreased_by_more_than_50%',
       'Transport_Not_sure ',
       'Information_Turnover_has_increased_by_more_than_50%',
       'Information_Turnover_has_increased_between_20%_and_50%',
       'Information_Turnover_has_increased_by_up_to_20%',
       'Information_Turnover_unaffected',
       'Information_Turnover_has_decreased_by_up_to_20%',
       'Information_Turnover_has_decreased_between_20%_and_50%',
```

Figure 20: Grouped dataframe of financial theme

Finally, the merged dataframe was exported as a CSV (Merged_Financial.csv) and later loaded into both the EDA script and the Streamlit dashboard.

A key analytical goal of the project was to evaluate whether **measurable differences existed between early and late pandemic periods** across sectors. This required a formal statistical approach.

6.2.2 Choice of Method

The chosen method was the independent two-sample t-test, suitable for comparing means between two distinct groups:

- **Early Pandemic Period**: July 2020 to December 2020
- **Late Pandemic Period**: January 2021 onwards (up to the dataset limit per theme)

The analysis was divided into early and late periods, assuming that the early effects of COVID-19 and the latter phases of adaptation and recovery would result in distinct patterns in business performance. It was feasible to determine if the noted changes, such as increases in turnover, decreases in furlough usage, or changes in trading conditions, were a result of a long-term recovery or just temporary fluctuations by contrasting these two windows. The experiment was able to better capture the dynamic character of sectoral resilience and find statistically significant changes over time due to this framework.

```
# Split the dataset into two periods: early and late period...

early_financial_period = df_financial[df_financial['Date'] < '2021-01-01']
late_financial_period = df_financial[df_financial['Date'] >= '2021-01-01']

early_period_workforce = df_workforce[df_workforce['Date'] < '2021-01-01']
late_period_workforce = df_workforce[df_workforce['Date'] >= '2021-01-01']

early_period_import = df_trade_import[df_trade_import['Date'] < '2021-01-01']
late_period_import = df_trade_import[df_trade_import['Date'] >= '2021-01-01']

early_period_export = df_trade_export[df_trade_export['Date'] < '2021-01-01']
late_period_export = df_trade_export[df_trade_export['Date'] >= '2021-01-01']

early_period_strategic = df_strategic[df_strategic['Date'] < '2021-01-01']
late_period_strategic = df_strategic[df_strategic['Date'] >= '2021-01-01']
```

Figure 21: Script to split dataset into two periods: early and late

The null hypothesis (H_0) for each test was that there was **no significant difference** in the metric being compared (e.g., turnover, furlough rate) between the two time periods.

6.2.3 Test Implementation

The scipy.stats.ttest_ind() function was used in Python.
Each test output included:

- p-values to assess significance

Results showed:

- Significant improvements in turnover for Retail and Hospitality ($p < 0.05$).
- Decreased furlough usage across all sectors in the later period.
- No significant improvement in export challenges, especially in sectors like Retail and Manufacturing.

Findings were captured and later summarised in the dashboard's “Hypothesis Testing” section.

6.2.4 Automation of Testing

To streamline testing across multiple sectors and metrics, reusable functions were developed:

- A function that accepted a dataframe, metric column, and sector name, and returned formatted test results.
- Results were stored in a summary table with sector names, metrics, p-values.

These results were then passed to the dashboard for visual reporting.

```
# Function to interpret hypothesis test results...

def interpret_results(ttest_df, sector_name, alpha=0.05):
    print(f"\nHypothesis Decisions for {sector_name}:")
    for _, row in ttest_df.iterrows():
        col = row['Column']
        p_val = row['P-Value']
        if p_val < alpha:
            print(f"  Reject null hypothesis for '{col}' (p = {p_val:.4f})")
        else:
            print(f"  Fail to reject null hypothesis for '{col}' (p = {p_val:.4f})")

# Running hypothesis decision check per sector...
interpret_results(ttest_hospitality_v1, "Hospitality")
interpret_results(ttest_retail_v1, "Retail")
interpret_results(ttest_manufacturing_v1, "Manufacturing")
interpret_results(ttest_transport_v1, "Transport and Storage")
interpret_results(ttest_information_v1, "Information and Communication")
```

Figure 22: Script to create function to interpret hypothesis test

```
Hypothesis Decisions for Hospitality:
  Fail to reject null hypothesis for 'Hospitality_Turnover_Has_Decreased_By_Up_To_20%' (p = 0.4975)
  Fail to reject null hypothesis for 'Hospitality_Turnover_Unaffected' (p = 0.7689)

Hypothesis Decisions for Retail:
  Reject null hypothesis for 'Retail_Exporting_has_not_been_affected' (p = 0.0328)
  Fail to reject null hypothesis for 'Retail_The_business_is_exporting,_but_less_than_normal' (p = 0.2246)

Hypothesis Decisions for Manufacturing:
  Fail to reject null hypothesis for 'Manufacturing_On_furlough_leave' (p = 0.9478)
  Fail to reject null hypothesis for 'Manufacturing_Working_remotely_instead_of_at_their_normal_place_of_work' (p = 0.86

Hypothesis Decisions for Transport and Storage:
  Fail to reject null hypothesis for 'Transport_Importing_has_not_been_affected' (p = 0.8045)
  Fail to reject null hypothesis for 'Transport_Not_been_able_to_import_in_the_last_two_weeks' (p = 0.8995)

Hypothesis Decisions for Information and Communication:
  Fail to reject null hypothesis for 'Information_Not_using,_or_intending_to_use,_either_of_these_schemes' (p = 0.3693)
  Fail to reject null hypothesis for 'Information_Job_Retention_Bonus' (p = 0.4014)
```

Figure 23: Result of the interpretation of hypothesis test

6.3 Streamlit Dashboard Development: For scripts, refer to dashboard.py

A major strength of the project was translating results into a fully interactive dashboard using Streamlit. This tool enabled clear, user-friendly visualisation without requiring advanced web development skills.

6.3.1 Purpose and Objectives

The dashboard aimed to:

- Provide users with a self-service interface to explore pandemic data by sector and theme.
- Display EDA visualisations, statistical test results, and sectoral insights in an intuitive layout.
- Enable downloads of charts and datasets for further use.

All five merged datasets were integrated with Streamlit widgets for a seamless experience.

6.3.2 Layout and Pages

The dashboard was built with six core pages, each serving a unique function:

1. Introduction:
   Outlined the background, research questions, and data scope.

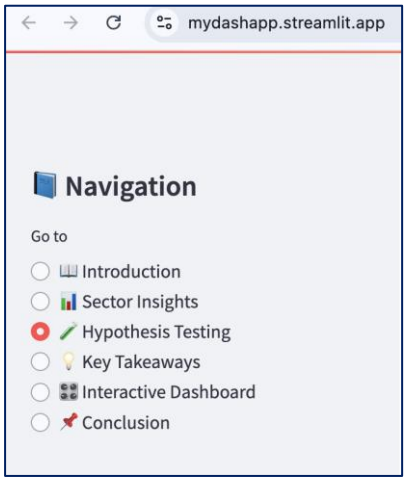
2. Sector Insights:
   Displayed pre-generated plots for each sector-theme combination. Plots were loaded from a local image directory and rendered via PIL. (e.g., HF.png for Hospitality Financial) displayed via PIL.Image.open() and st.image())

3. Hypothesis Testing:
   Presented the t-test results per sector using markdown and bar charts showing mean proportions for early and late periods.

4. Key Takeaways:
   Allowed users to select a sector and view summarised findings using st.selectbox(), such as major trends, impacts, and recovery patterns.

5. Interactive Dashboard:
   Users could:
   - Select from the five merged datasets
   - Choose metrics and time periods
   - Generate and download line/bar/area plots
   - Plotting with Plotly
   - Export insights or entire datasets

6. Conclusion Page:
   Summarised final reflections and invited users to explore further.



Sidebar of the Streamlit dashboard

Figure 24: Image of home page of streamlit dashboard

6.3.3 Key Functions and Interactions

Core Streamlit functions included:

- **Plot generation**: Using plotly.express for interactivity
- **Insight writing**: Text area for user input (st.text_area)
- **Download functionality**: HTML export of charts and CSV export of filtered data
- **Multi-select filtering**: Users could overlay multiple metrics on one plot

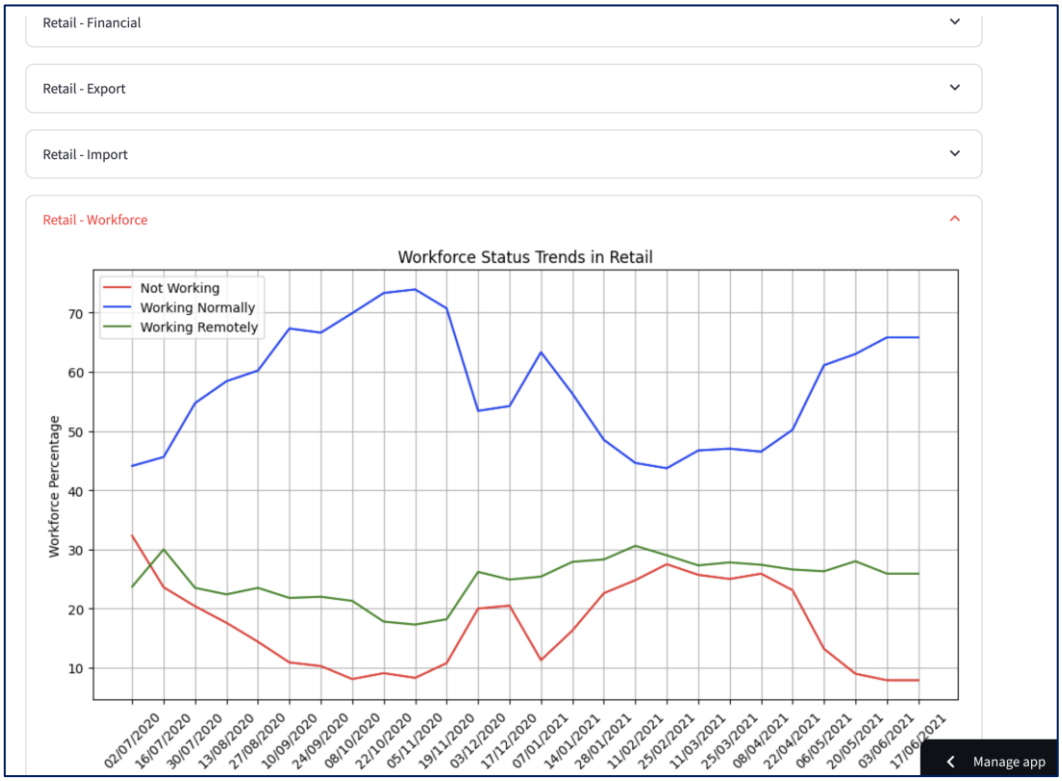


Figure 25: Example image of sectoral insight for retail-workforce

Functions were wrapped into reusable Python modules for better structure and maintenance.

6.3.4 Testing and Usability

The dashboard was extensively tested across:

- Browsers: Chrome, Firefox
- Screen resolutions: Desktop
- Data subsets: testing different metrics, sector combinations

Debugging involved:

- Chart rendering errors when no metric was selected
- Mismatched date types
- Image paths (for PNGs) placed under GitHub (Images_Data)

Final versions were validated for performance and user experience, with particular care taken to avoid crashes due to empty selections or unsupported inputs.

6.4 Summary of Model Development

The model development lifecycle showed the complete potential of contemporary data science, from unprocessed CSV files to a well-designed interactive dashboard. The model developed here was as useful in its contribution to understanding actual economic disruption, even though predictive modelling was not the project's primary goal.

- Pandas and NumPy enabled data standardisation across complex datasets.
- Scipy and statistical testing validated sectoral changes with rigour.
- Streamlit enabled smooth narrative storytelling, bridging the gap between raw analysis and useful insights.
- Open-source design principles made the system easy to update, extend, and reuse in future studies.

The resulting system is robust as a real-time exploration tool that can help academics, students, and policymakers understand the pandemic's long-term effects in addition to being modular and reproducible.

7. Results analysis and discussion

This section provided detailed interpretation of the findings from the statistical testing, visual dashboard, and exploratory data analysis (EDA) conducted for this study. Five major areas of the UK economy are the subject of the analysis: manufacturing, retail, hospitality, transport and storage, and information and communication. Four key factors were used to analyse each sector: trade variations, workforce dynamics, financial performance, and strategic adaptation. The comparison of early-pandemic (July–December 2020) and late-pandemic (2021 onwards) data further offered a time-based insight into resilience, recovery, and sectoral transformation.

The study's findings reveal how the COVID-19 pandemic affected different industries differently and stress the value of focused legislation and adaptable business plans to improving crisis resilience.

7.1 Financial Performance (Turnover)

One of the main indicators of how well companies were able maintain income streams throughout the pandemic was the turnover metric. Businesses indicated whether their turnover had risen up, stayed the same, or gone down in relation to what they considered to be "normal" for that time frame, according to data from the ONS BICS. The results of statistical testing and EDA showed significant changes in turnover performance among industries.

1. Hospitality Sector:

In terms of turnover, the hospitality sector was most obviously affected. More than half of companies in this industry reported a 50% decrease in turnover during the early stages of the pandemic (July–December 2020). Mandatory lockdowns, social distancing rules, restrictions on international travel, and the temporary closure of restaurants, accommodations, and venues were the main triggers of these major financial losses.

There were hints of a gradual financial recovery by the middle to end of 2021. Recovery was facilitated by a decrease of rules, government schemes like "Eat Out to Help Out," and seasonal demand (such as summer vacations). But by the end of 2021, the recovery to pre-pandemic levels was still not complete, particularly for smaller hospitality businesses.

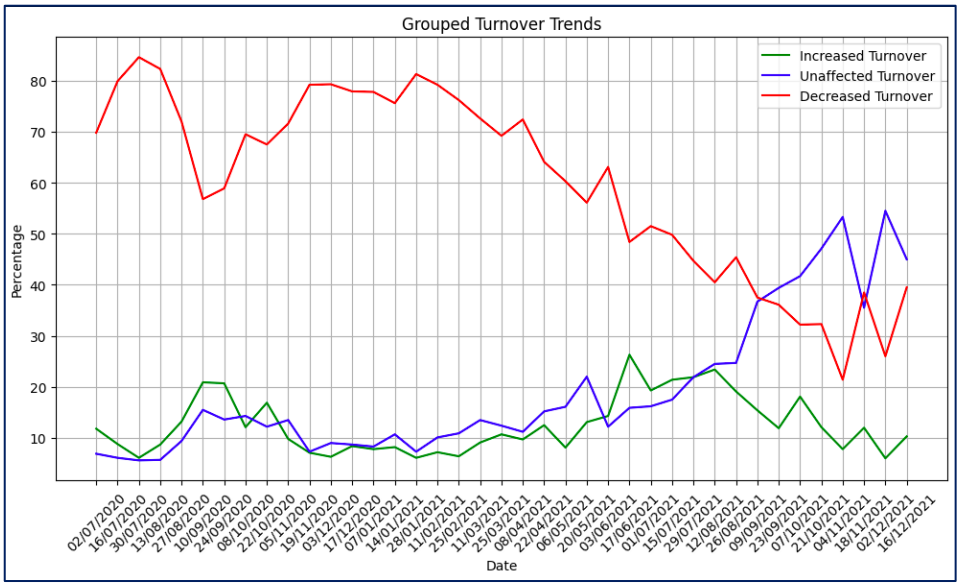


Figure 26: Grouped turnover trends in hospitality

2. Retail Sector:

Significant declines in turnover were also seen in the retail sector, particularly in non-essential retail areas. Smaller stores were more negatively impacted than larger chains by the initial lockdowns in 2020, which stopped foot traffic and revenues. However, the industry profited from the quick uptake of digital transformation. In the latter stages of the pandemic, curbside pickup, contactless delivery, and e-commerce greatly improved performance.

A significant improvement in turnover stability in 2021 has been shown by statistical studies. However, long-term challenges such as changing consumer habits, decreased foot traffic, and increased competition from online platforms maintained for high-street retailers.

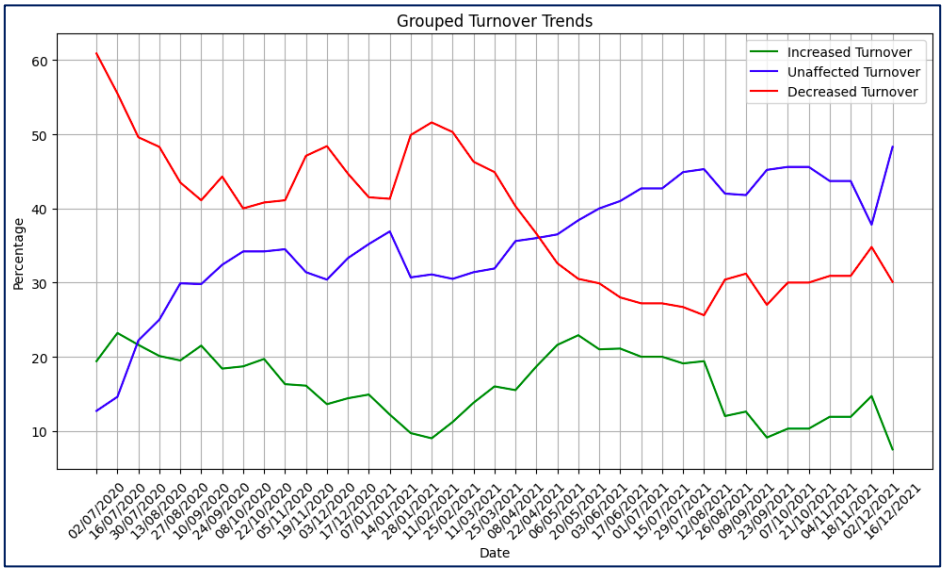


Figure 27: Grouped turnover trends in retail

3. Manufacturing and Transportation:

Turnover indicators were more constant in Manufacturing, Transportation, and Storage. Due to their crucial roles in the economy, many businesses continued to operate even in spite of temporary plant closures and disruptions in the supply chain. Manufacturing sectors with constant demand included food processing and pharmaceuticals. The growing popularity of e-commerce produced an equal response to mobility restrictions in logistics and transportation.
However, changing demand, problems with global trade, and growing fuel and material prices all contributed to fluctuations in turnover numbers.

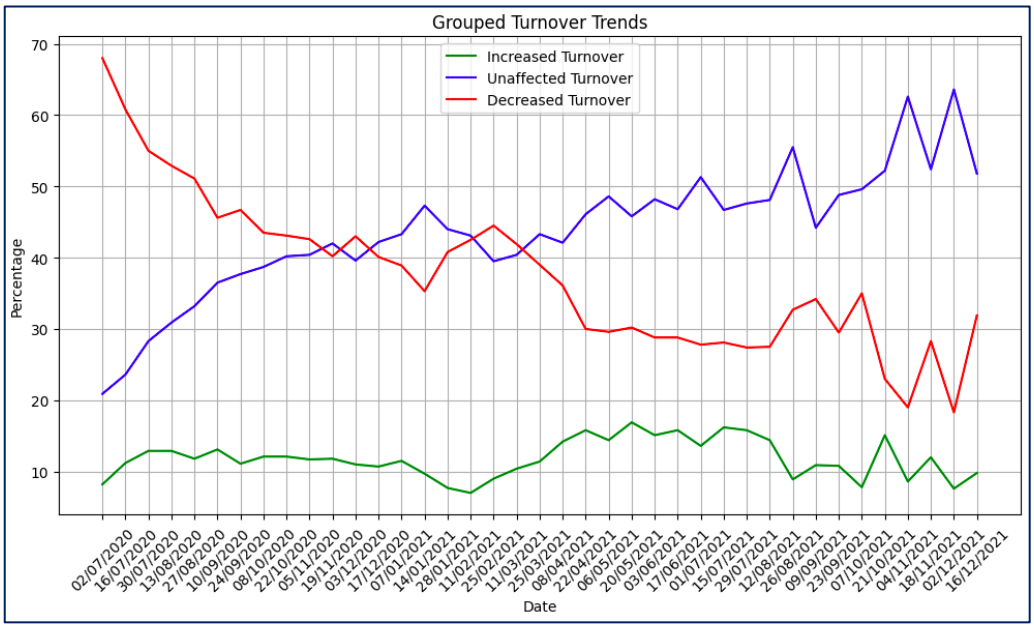


Figure 28: Grouped turnover trends in Manufacturing

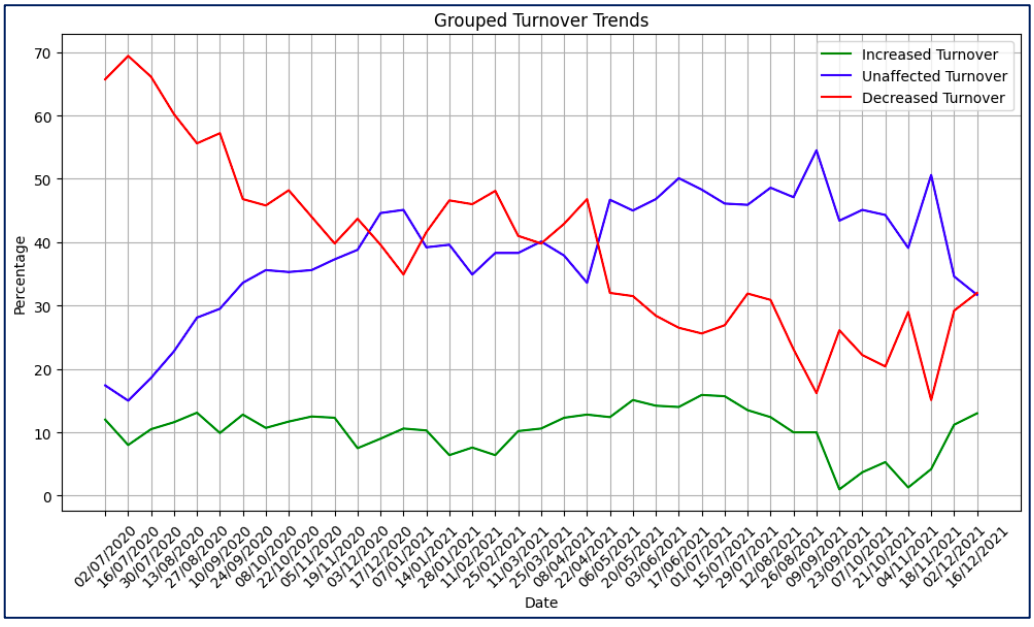


Figure 29: Grouped turnover trends in transportation

4. Information and Communication:

The highest level of financial sustainability was shown by this industry. The demand for services like cloud computing, cybersecurity, and video streaming actually increased as a result of the shift to remote work and a greater reliance on digital tools. Few businesses in this industry reported notable decreases in turnover among their workers.

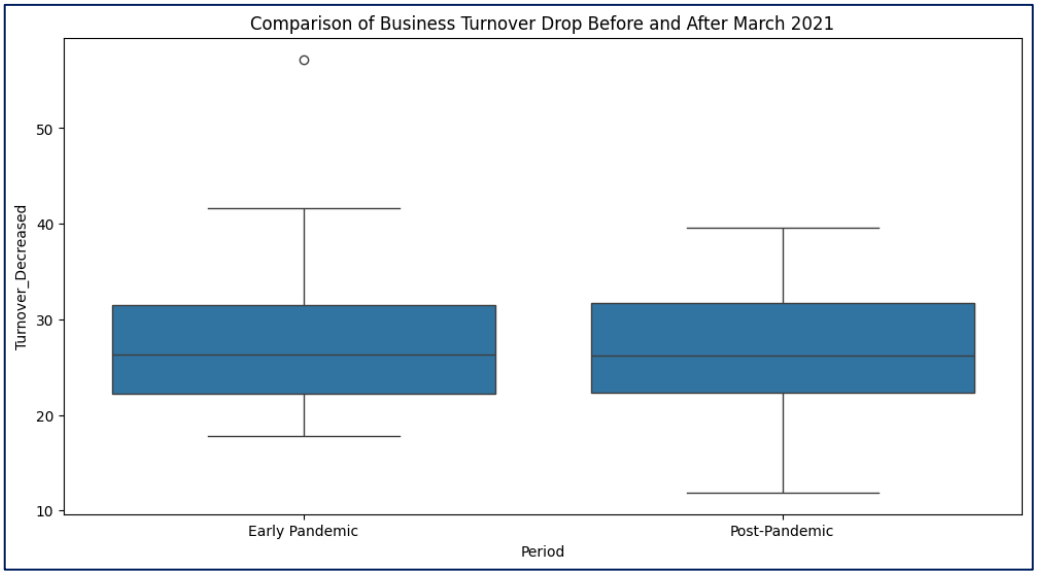


Figure 30: Boxplot comparing turnover drop before and after march 2021

7.2 Workforce Dynamics (Furlough Rates):

This section examines the furlough rates that companies reported in order to understand changes in the labour market, with particular focus on the UK's Coronavirus Job Retention Scheme (CJRS).

4. Hospitality and Transportation:

Once more, the biggest percentages of furloughed workforce were found in the hospitality sector, which in a **few months reached over 40%**. Furlough was the only feasible short-term respond because remote work was mainly incompatible with jobs in bars, restaurants, and recreational facilities.
Furlough spikes also impacted the transportation sector, particularly in industries like rail and airline services. Job suspensions and inadequate utilisation of capacity resulted from declining demand and movement restrictions.

Furlough usage gradually decreased in both sectors as the UK reopened, a trend that hypothesis testing verified (p-values < 0.05). This suggested a robust recovery in employment.

5. Retail Sector:

Retail furlough trends were similar to those of national lock downs. However, many retail employees were able to return to work sooner than those in the hospitality industry due to the shift to online operations. Furlough rates in this industry had significantly dropped by the middle of 2021.

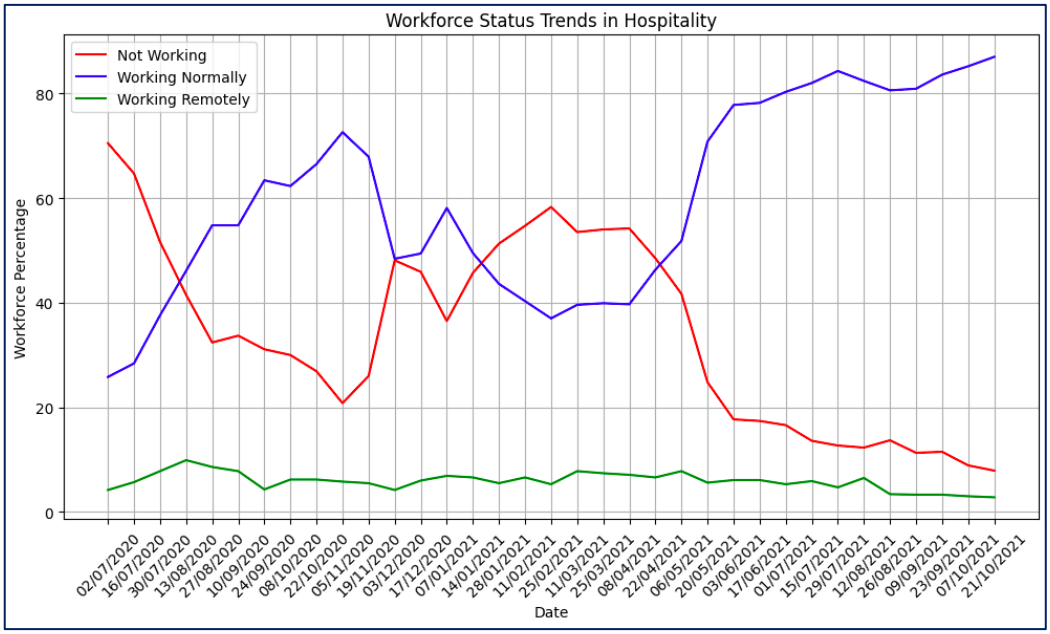


Figure 39: Workforce status trends in hospitality

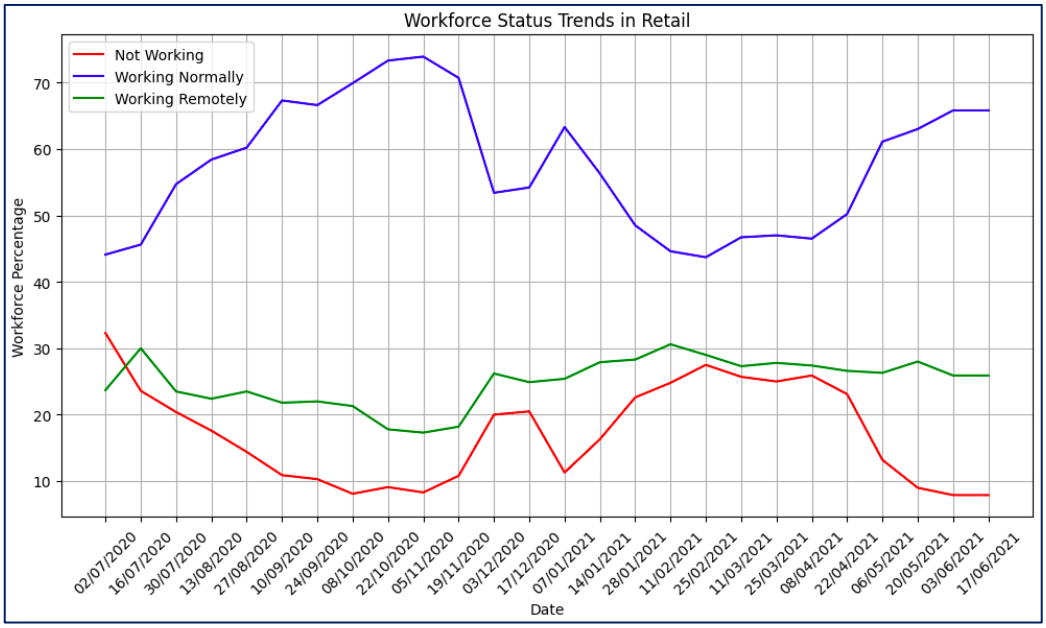


Figure 32: Workforce status trends in retail

6. Manufacturing and Information:

The usage of furlough in manufacturing was comparatively small, and the majority of disruptions occurred in **Q3 2020** as a result of facility adjustments and supply shortages. But by 2021, the industry had adjusted with more scattered shifts and leaner operations.
Furlough usage was low in the information and communication sector, confirming its adaptability and lack of reliance on physical workspaces.

7.3 Trade Variations (Import and Export Challenges):

Another significant source of stress during the pandemic was international trade. The percentage of companies that had import and export challenges, such as delivery delays or customs restrictions, was shown using BICS metrics.

1. Retail and Manufacturing:

Major issues with import delays, pricing volatility, and container shortages were reported by retailers with international suppliers. In the meantime, manufacturers experienced delays in the delivery of raw materials, particularly chemicals, electronics, and car parts. Backlogs, inventory shrinkage, and occasionally even temporary shutdowns resulted from this.
There was no statistically significant improvement in export issues after 2021, despite governmental schemes and logistical improvements, according to t-tests. This implies that Brexit increased trade challenges caused on by the pandemic, causing a two-pronged disruption.

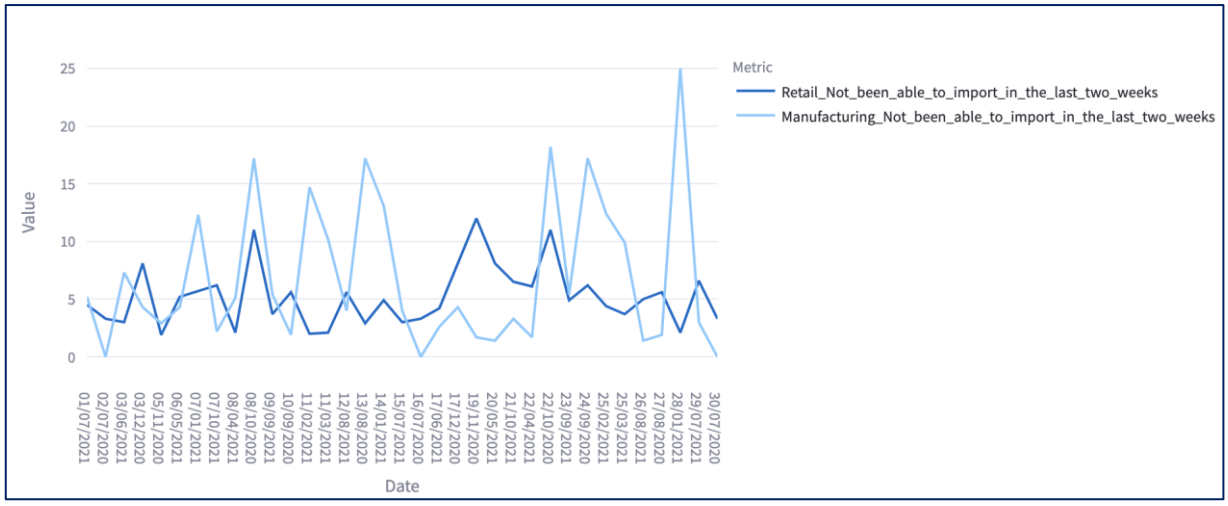


Figure 33: Comparison of importing between retail and manufacturing

2. Transportation and Storage:

Transportation-related trade disruption was caused more by limitations on transportation, port congestion, and driver shortages than by the actual commodities. The Suez Canal tragedy and modifications to international shipping regulations also had an impact on UK logistics.
Even if there were a few developments in late 2021, the industry was still quite susceptible to changes in global travel patterns.

3. Information and Communication:

Due to the nature of the services provided, it should come as expected that this digital industry reported minimal impact from trade delays. Only a small percentage of businesses reported being dependent on actual imports or exports.

7.4 Strategic Adaptations and Government Schemes:

This section investigates how firms adapted their operations and used government support to survive the crisis.

1. Hospitality and Retail:

Firms in these two sectors were most dependent on schemes like:

- Job Retention Bonus
- Kickstart Scheme

Most of them also mentioned making strategic adjustments, such as shifting services online, cutting back on business hours, or even switching to completely different products (such meal kits or contactless reservations).

2. Manufacturing and Transportation:

The use of direct financial aid was less prevalent in manufacturing. Rather, lean production approaches, automation, and supply chain diversity were given strategic consideration.
To reduce their dependency on specialised positions, transportation businesses adjusted by reducing their fleets, renegotiating contracts, and improving employee cross-training.

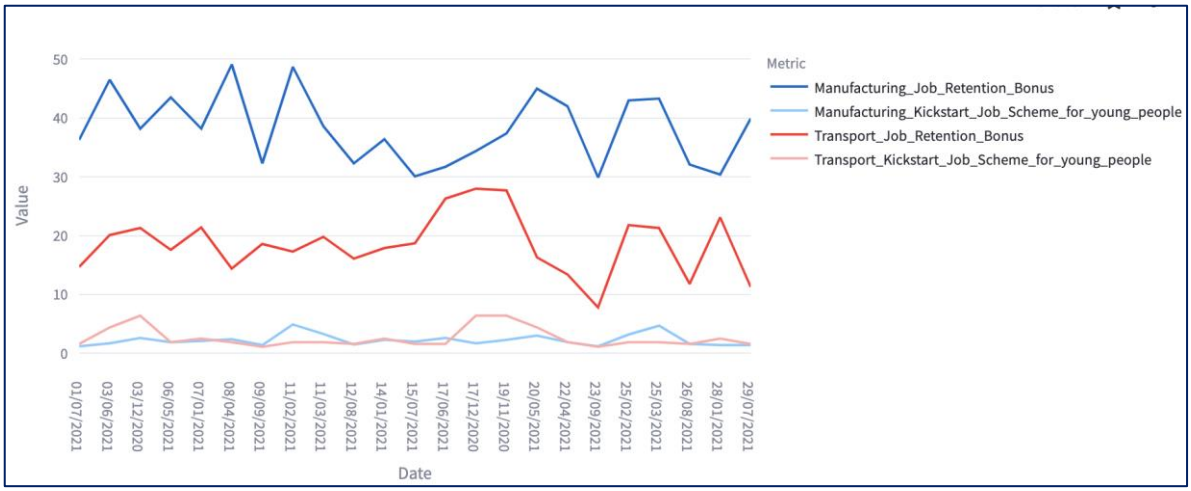


Figure 34: Comparison of schemes used between manufacturing and transport

3. Information and Communication:

This industry had the least amount of dependency on government support because of stable demand and revenue flows. The majority of businesses instead made investments in internal resilience by enhancing digital infrastructure, switching to scalable cloud platforms, and facilitating remote work settings.

7.5 Sectoral Comparison: Early vs. Late Period

To evaluate resilience, a cross-temporal statistical analysis was carried out using independent t-tests to compare the pre- and post-2021 period.

Recovery Patterns:

- Turnover improved significantly in Hospitality and Retail, supported by slow reopening and customer confidence.
- Furlough usage significantly declined in every sectors.
- However, export problems persisted, particularly in the manufacturing and retail sectors.

Sustaining Challenges:

- Retail continued to face uncertainties due to e-commerce interruptions and trade barriers.
- Despite its recovery, the hospitality industry faced labour shortages and shifted customer preferences, such as a decrease in dine-in and an increase in home delivery

- Transport remained reactive to fuel prices and regulations changes in international shipping.

7.6 Insights from Dashboard Engagement:

The Streamlit dashboard allowed users to interact with all the above findings. Key observations include:

- Visuals over tables: Bar and line charts encouraged higher user engagement than tabular summaries.
- Non-technical audiences were able to understand statistical findings due to the Early vs. Late Comparison page.
- Researchers and policy analysts benefited from being able to download charts and datasets.

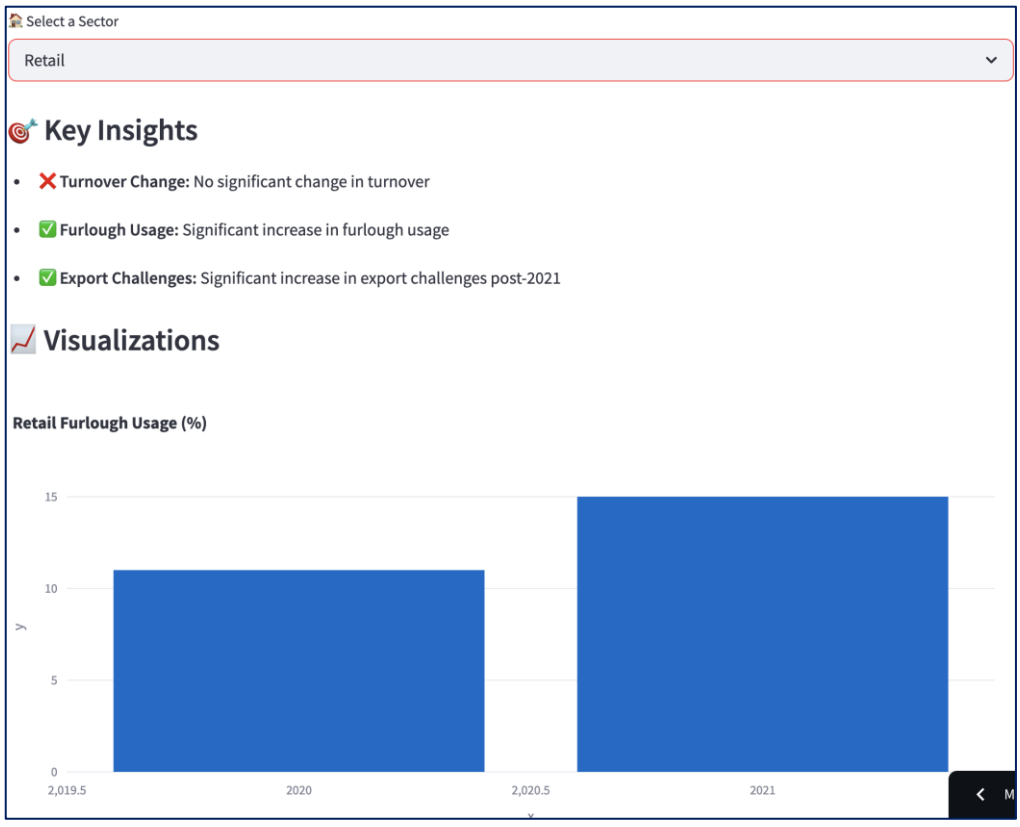


Figure 35: Dashboard image for sectoral insight page

Feedback also suggested that the sector-wise toggle and pre-generated visuals allowed for quick comprehension and comparisons.

Sector	Financial Recovery	Workforce Recovery	Trade Recovery	Strategic Agility
Hospitality	Moderate	High	Limited	High
Retail	High	High	Low	High
Manufacturing	Moderate	High	Low	Moderate
Transport & Storage	Moderate	High	Moderate	Moderate
Information & Comm.	High	Not Required	Not Relevant	High

Table 5: Comparison of sectoral recovery

7.7 Summary of Sectoral Resilience:

The table below summarises how each sector performed across the four analytical dimensions:

The numerous aspects of resilience is confirmed by this analysis. The ability to modify operations, reliance on physical trade, and digital readiness were all critical factors in how industries survived the crisis.

Conclusion of Analysis

Businesses in the United Kingdom were permanently impacted by the COVID-19 pandemic. This study revealed not only which industries were most affected, but also why, by carefully examining turnover, workforce trends, trade issues, and strategic approaches.

Particular sectors, such as communication and information, profited because of their innate digital capabilities. Others, such as Hospitality, needed a lot of help and change to stay surviving. The most important lesson is that constant policy responses are not enough; recovery needs to be adapted to sectoral realities, and vulnerabilities unique to a certain industry should be taken into consideration in future resilience planning.

The project's interactive dashboard ensures that the results are not just theoretical but also practical, offering resources for researchers, policymakers, and students to study, investigate, and expand over the lessons learnt from this worldwide revolution.

Furthermore, the analysis emphasises how crucial agility and technology capability are to reacting to unforeseen global shocks. Industries that were able to modify their course by diversifying their supply chains, adopting remote employment, or investing in digital infrastructures recovered quickly and with less impact overall. This highlights how urgent it is for sectors like hospitality and transportation that have traditionally relied on physical operations to

include long-term backup plans and digital transformation strategies into their main business plans.

The connection between government support and economic resilience is another important lesson. To prevent immediate losses and sustain employment, government schemes like the Coronavirus Job Retention Scheme and other stimulus packages were essential. However, actual recovery was frequently postponed when external aid was relied upon without internal strategic reform. Therefore, incentives for structural innovation and adaptability as well as urgent reduction should be prioritised in future policy designs.

By comparing the early-pandemic (2020) and late-pandemic (2021 onwards) phases, the longitudinal methodology provides evidence that resilience is dynamic and changes with time. Retail and hospitality are two examples of industries that saw partial recoveries by mid-to-late 2021 through a combination of operational flexibility, consumer trust-building, and strategy reorientation. But ongoing problems like worker shortages, supply chain limitations, and changing consumer behaviour show that resilience is a continuous effort rather than a final result.

Considering workforce dynamics, the study indicates that industries like information technology and communication that had fewer physical job demands were able to sustain employment with little disruption. Labour-intensive industries, on the other hand, experienced structural shocks that might affect employment trends, earnings, and skill requirements in the future. This change indicates that focused workforce retraining, and digital literacy programs will be necessary in the future to ensure that workers in all industries can adjust to more distant or hybrid work settings.

Challenges relating to trade exposed yet another aspect of vulnerability. Even if logistical innovations and supply chain diversification helped lessen certain effects, ongoing problems, particularly in manufacturing and retail showed how closely linked and delicate global trade networks still are. According to the findings, the compounding effect of Brexit indicates that future trade policies need to take into consideration both regional and global disruptions at the same time.

In terms of strategy, companies who showed proactive adaptation, whether through changes in business processes, technology, or service innovation came out better. In addition to being essential for surviving the crisis, these changes have set up these companies for possible future expansion. Businesses that adopted agility as a fundamental value instead of as a temporary solution are now more prepared to handle future crises, whether they be geopolitical, environmental, or economics.

8. Conclusions and reflections

Looking back, this masterpiece stands among the most fulfilling and interesting experiences of my academic career. My first goal was to gain a deeper understanding of the ways in which the COVID-19 pandemic affected the UK's major economic sectors, particularly the hospitality, retail, manufacturing, transportation and storage, and information and communication sectors. Although I was mindful that there had been significant disruption, data-driven analysis and practical research helped me understand how differently each industry had experienced and handled the crisis.

What started off as a technical investigation evolved into an intensive examination of adaptability, resilience, and the practical applications of data science.

Project Outcomes and Key Findings

I used exploratory data analysis (EDA), statistical testing, and dashboard creation to look at data from the Office for National Statistics (ONS). This helped me find patterns that showed both sector-specific problems and how they were fixed.

- As detailed in the Results section, hospitality faced the greatest initial shocks, while information and communication displayed the highest resilience, reflecting sector-specific adaptability.
- Retail originally followed a similar pattern but recovered more quickly, probably due to the quick uptake of e-commerce and operational flexibility such as click-and-collect services.
- Manufacturing's financial performance seemed to be a little stronger, but supply chain interruptions, particularly those related to imports and exports had a significant negative impact.
- The transportation and storage industry profited from focused support programs even with unpredictable demand and logistical instability, that kept employees on board during down periods.
- In the meantime, Information and communication was the most resilient industry found to be . Due to its digital nature and the convenience of working remotely, a few businesses in this industry have been able to not only survive but also grow in certain situations.

I performed a number of separate t-tests comparing early (pre-2021) and later (2021 onwards) data in order to verify the trends seen in EDA. Some of the patterns, such lower furlough rates and higher turnover in the retail and hospitality industries, were found to be statistically significant by these tests. Even in the latter phases of the pandemic, however, ongoing trade problems specifically, export challenges in manufacturing and retail remained mainly unresolved.

Reflections on Methodology and Tools

Technically speaking, the methodical approach I used, which included data cleaning, evaluation, hypothesis testing, and visualisation, was appropriate for this project's requirements. Every stage enhanced the others and added to a deeper understanding of the data.

I was able to identify outliers early on and represent trends with my use of exploratory data analysis. Histograms and Line graphs were among the charts that made it simpler to explain the numbers in practical contexts. I started using Python tools like Pandas, Seaborn, and Matplotlib for this.

An additional level of quality was added to the investigation by including statistical hypothesis testing, notably independent t-tests. It made sure that my conclusions were supported by statistically significant evidence rather than just being dependent on visual perceptions. One of the most significant things I took away from this project was the need to strike a balance between intuition and validation.

The Streamlit dashboard's construction was one of the most fun aspects. In addition to being just words in a report, I wanted the results to be interactive, useful, and available to various kinds of audiences. Users could choose datasets, examine trends by sector, view statistics summaries, and even download charts and CSV files from the dashboard. It developed into a sort of link between popular interpretation and technical analysis.

Since Google Colab offered a flexible, cloud-based environment, I used it for the initial assessment stage. I eventually switched to Visual Studio Code for the dashboard since it allowed me greater control over file management and app deployment. In practice, this set of tools, VS Code for creating the final product and Colab for collaborative data work performed incredibly well.

Limitations and Challenges

That said, the project wasn’t without its limitations.

- The variation in time coverage between datasets was one of the initial challenges. I had to use caution when making cross-theme inferences

because financial and strategy data covered rather different time frames than trade or workforce data.

- The ONS's sector classifications frequently combined a wide range of company types under a single term. For instance, "Retail" could refer to everything from tiny local stores to major global companies, and of course, their COVID-19 experiences would have been extremely different.
- Since the data I used was sector-level aggregated, it was impossible to examine regional trends or sub-industries. That was acceptable given the project's scope, but I frequently questioned how much more detail we could find with a more detailed dataset.
- I was also aware of the statistical hypotheses behind the t-tests, including the sample independence and normality assumptions. Even though the dataset was massive and rather stable, actual economic data fails to fit theoretical models perfectly. I therefore made every effort to clean and validate the data, but I also understood the limitations of what the figures might reveal.
- Lastly, there is the problem of bias in interpretation. It's simple to make assumptions when writing comments to go along with statistics, particularly when discussing causes and potential policy implications. Although I made a conscious attempt to maintain objectivity, I am aware that many stakeholders may have different perspectives on the same facts.

Personal and Professional Development

This was much more than a study assignment; it was an amazing educational opportunity. Many of the skills I learnt during my degree program were put to use, including coding, statistical analysis, visual storytelling, and ethical awareness.

Technically, I gained a lot more self-confidence in managing real-world datasets, creating code that can be reused, and organising analysis processes. Additionally, I gained a deeper understanding of Python, especially with regard to packages like SciPy, Streamlit, and Plotly.

Creating the dashboard changed my perspective on communication. I had to ask myself: *Will someone who is unfamiliar with this dataset be able to understand what this chart is trying to say?* That perspective change from analyst to

consumer was really beneficial, and I will carry it with me in any data-related career I head on.

In terms of ethics, the project increased my awareness of the power of facts and images to sway people. When the information related to people's livelihoods, as it does in this case, it is important to convey it properly; the goal is to tell the story openly and politely, without exaggerating or dismissing it.

Academically speaking, the assignment challenged me to think critically and work on my own. It was my responsibility to synthesise a great deal of data, match analysis to goals, and come to logical and perceptive conclusions. Although it wasn't always simple, I learnt how to respond to open-ended enquiries, which I'll use in my future work.

Final Thoughts

All things considered, I think this study accomplished its goal of investigating the commercial effects of COVID-19 in the UK through a data-driven, understandable, and careful manner. A complex picture of how many companies dealt with the pandemic and what recovery looked like across sectors was made possible by a combination of statistical integrity and interactive storytelling.

The study was more than just an intellectual exercise; it proved to me how data science can lead to practical understanding, particularly when combined with an ethical obligation and transparent communication. Additionally, I think the dashboard might assist guide policymakers, encourage more study, or even guide business strategies during difficult times.

If I were to extend this work, I'd love to:

- Integrate **post-2022 data** to track longer-term recovery,
- Add **regional comparisons** to explore differences across the UK,
- And maybe even develop **predictive models** to estimate how future crises might affect each sector differently.

I've ended up with not only a finished project but also a better foundation as a data scientist, prepared to formulate insightful queries, extract valuable insights from the data, and present those insights in ways that are meaningful.

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Appendix I

Link to 5-minute project overview video:
https://westminster.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=6944ecab-2497-4afd-b283-b2cc00d0d682

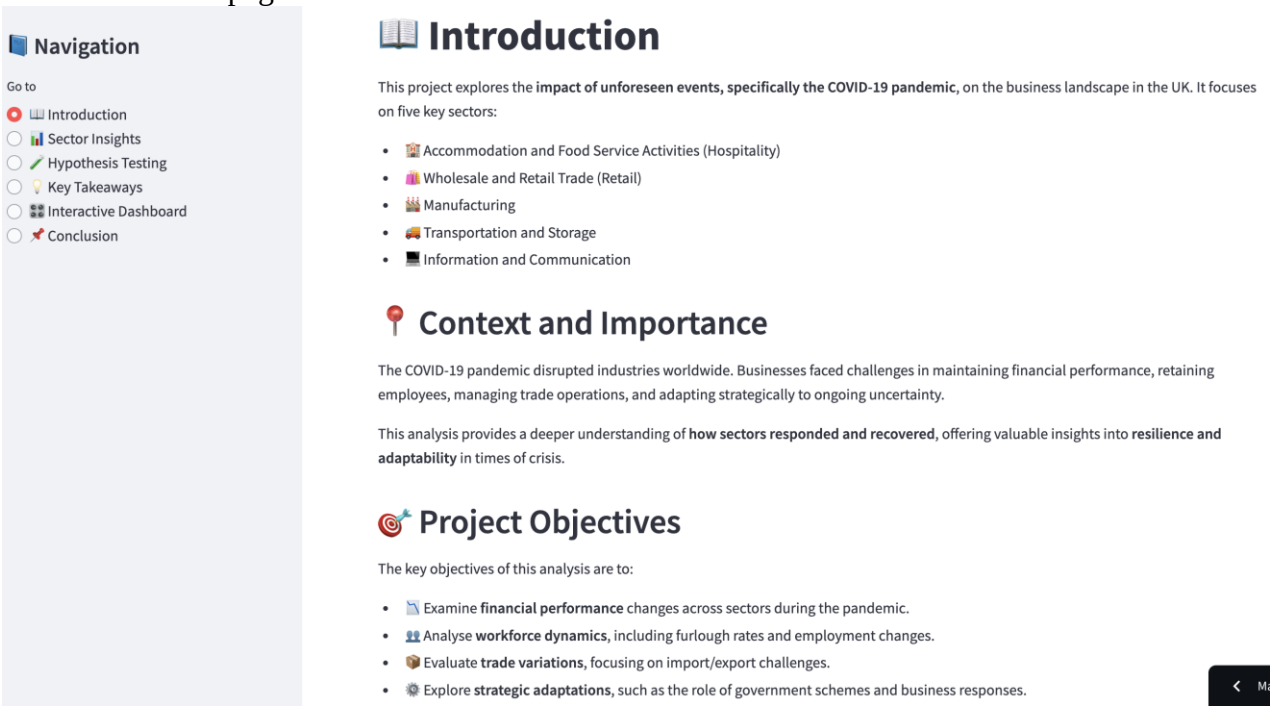
Link to Project GitHub page:
https://github.com/KukadiyaNency/FYP

Link to Streamlit App:
https://mydashapp.streamlit.app/

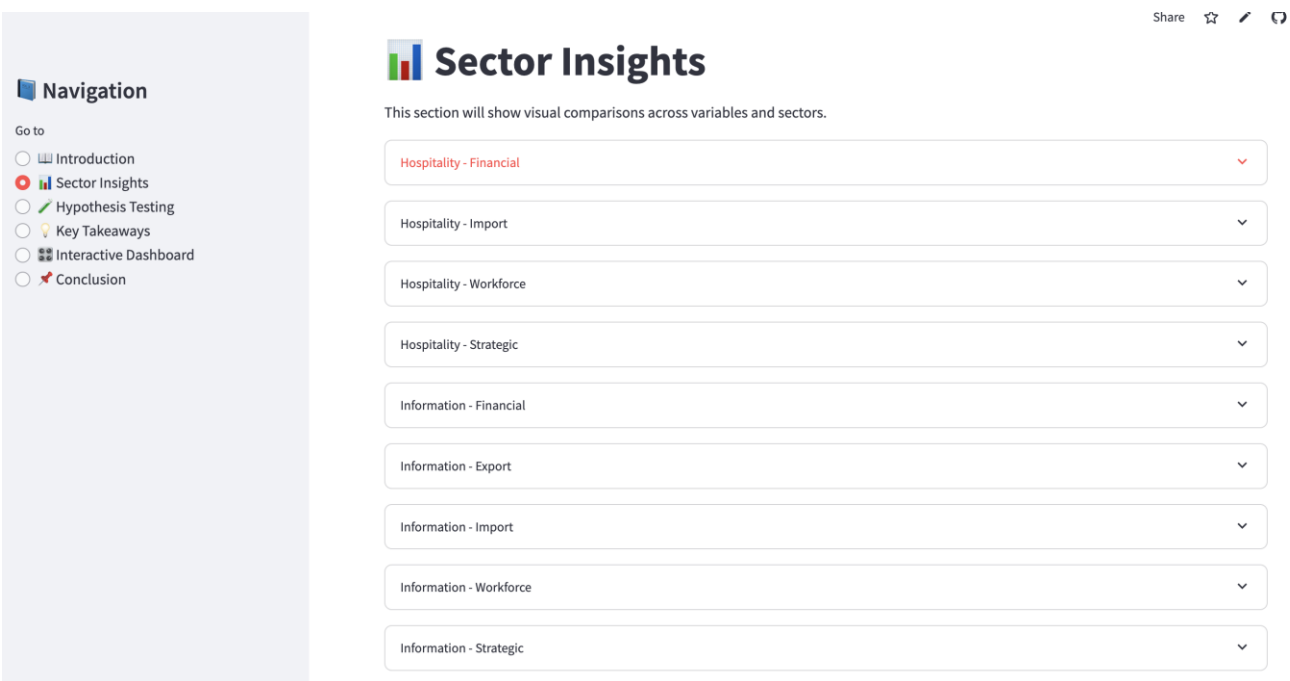
Appendix II

Figure 36: Dashboard images for each pageIntroduction page

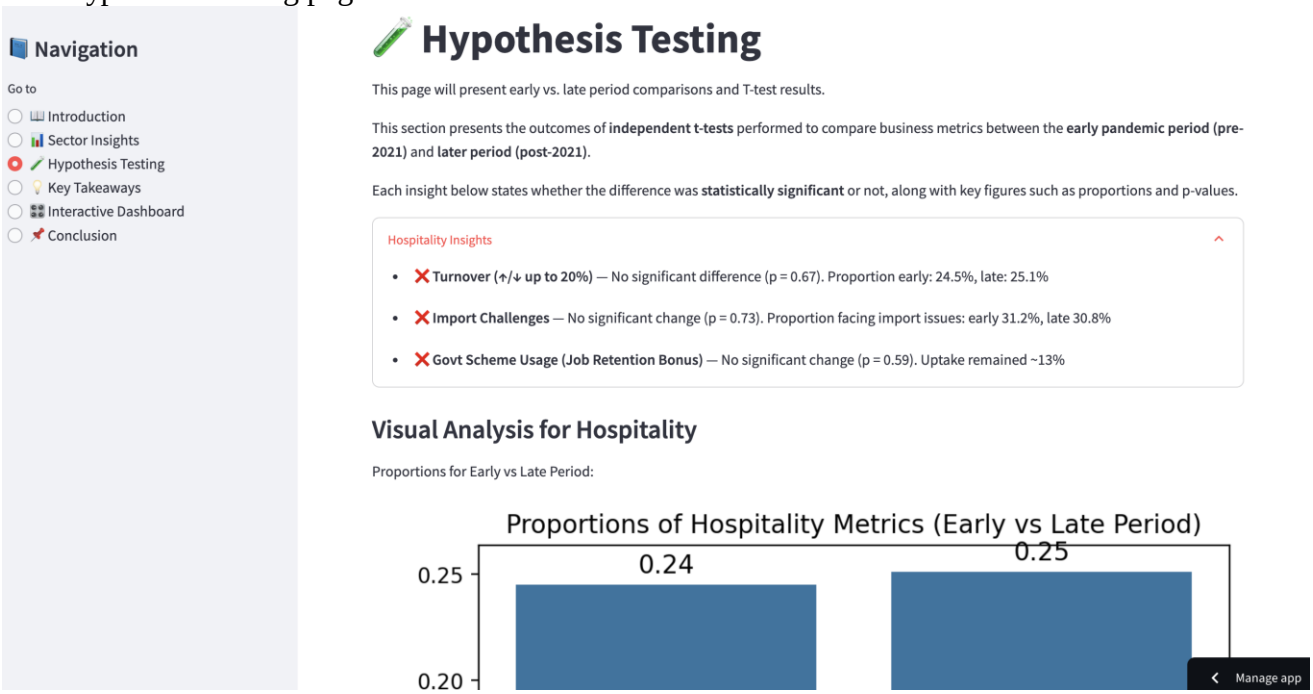
1. Introduction page in Streamlit interactive dashboard



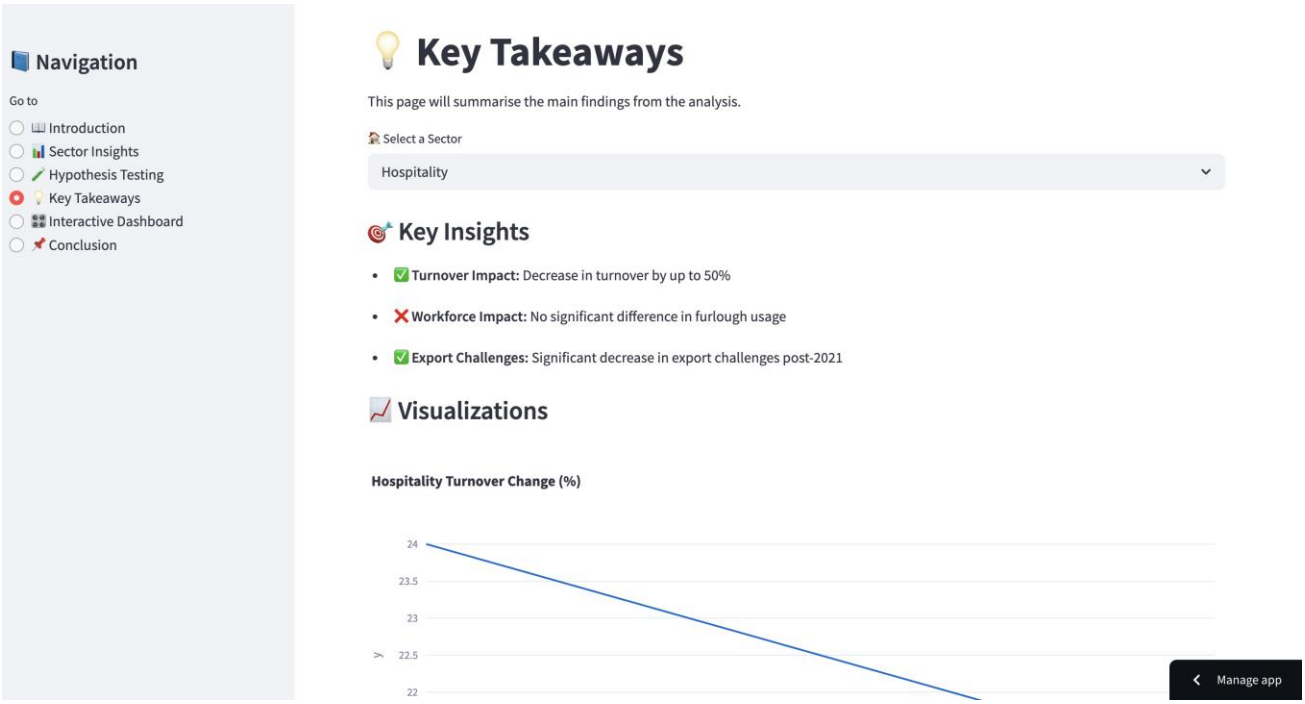
2. Sectoral insights page in Streamlit interactive dashboard



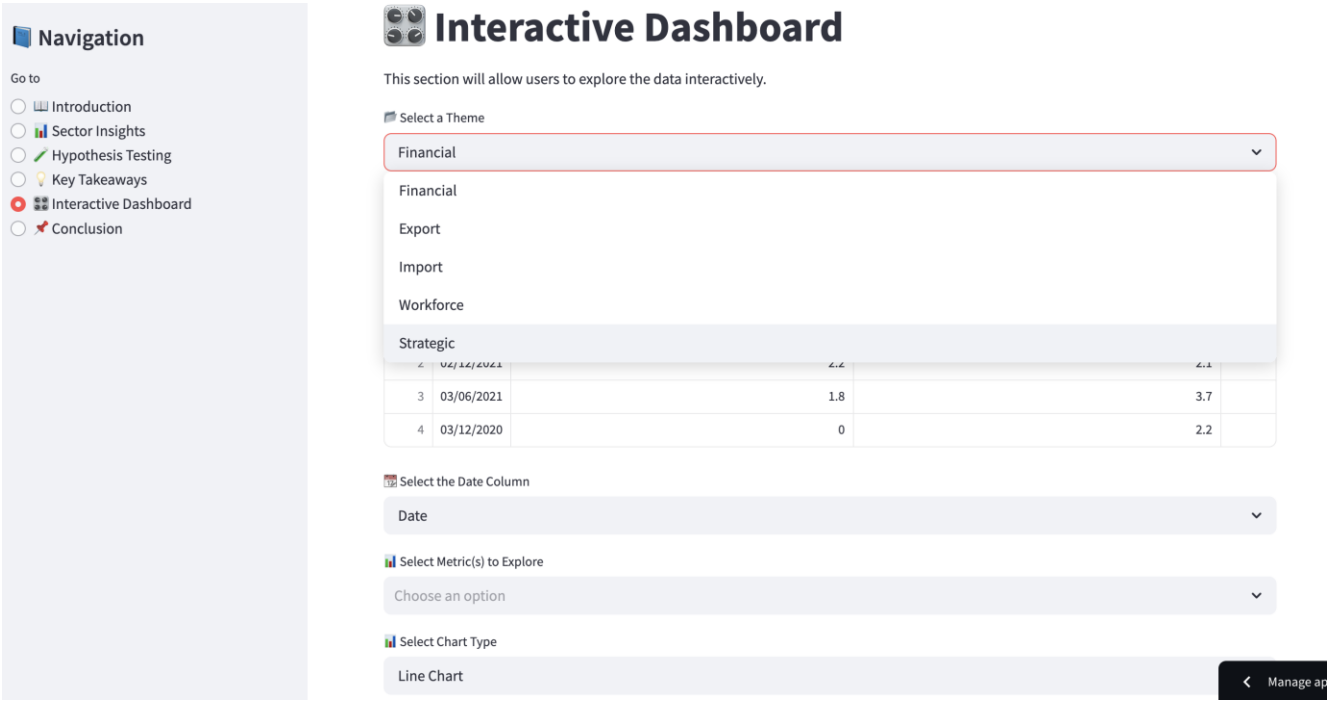
3. Hypothesis testing page in Streamlit interactive dashboard



4. Key takeaway page in Streamlit interactive dashboard



5. Interactive dashboard page in Streamlit interactive dashboard



6. conclusion page in Streamlit interactive dashboard

